

Monopolistically Skewed Business Cycles: On the Origins of Distributional Fluctuations^{*}

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September 14, 2026

Abstract

How do aggregate shocks jointly move mean growth, dispersion, and skewness over the business cycle? In U.S. public-firm data, these three moments fluctuate in stable patterns, yet with cross-sectional heterogeneity: larger firms have lower, more stable growth dispersion but substantially larger skewness fluctuations. We explain these cyclical dynamics and their *size gradient* through heterogeneous exposure and demand curvature. In general homothetic demand systems, a strengthened form of Marshall's second law makes firms' equilibrium policies from cost shocks to log real revenue concave, generating skewed growth. Under mild conditions, larger firms have flatter policies with stronger curvature, rationalizing the size gradient. We confirm this empirically by jointly recovering relative policy slopes, curvature, and idiosyncratic volatility across firm sizes from growth variance and skewness dynamics. Firm-specific responses to six independently identified aggregate innovations broadly reproduce the size gradient in growth moments. Using measured firm-level COVID exposures, we further link aggregate shocks to distributional growth asymmetry.

Keywords: business cycle, skewness, demand curvature, shock propagation

JEL Codes: E32, D21, L11

^{*}This paper supersedes *Granular Responses to Aggregate Shocks: An Explanation of Skewness over the Business Cycle* by Niklas Schmitz. We thank Guido Ascari, Igli Bajo, Anmol Bhandari, Florin Bilbiie, Leonard Bocquet, Paul Bouscasse, Vasco Carvalho, Elisa Faraglia, Alessandro Ferrari, Lukas Freund, Maren Froemel, Xavier Gabaix, Chryssi Giannitsarou, Fatih Guvenen, Sean Holly, Felix Kübler, Hanbaek Lee, Glenn Magerman, Shane Mahen, Leonardo Melosi, Mathis Momm, Charles Parry, Alba Patozi, Nicola Pavoni, Margit Reischer, Petr Sedlacek, Adam Hal Spencer, and Philipp Sternal for their valuable feedback and participants at the 2025 EWMES, 2026 AMES (CSW), 2026 EMES, and 2025 XAMsterdam conferences, the 2026 Barcelona Summer Forum, the Cambridge Macro PhD Workshop, and seminars at De Nederlandsche Bank for helpful comments. We also thank Stephen Hansen for generously sharing data with us.

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I Introduction

Are higher moments of firm growth primitive features of business-cycle shocks, or do they emerge from firms' propagation of aggregate disturbances? This paper develops the latter view. We show that complex distributional business-cycle dynamics in the cross-sectional mean, dispersion, and skewness of growth can emerge from a common aggregate shock and just two ingredients: firms differ in their exposure to the shock, and their log real-revenue policies are concave in this exposure-weighted shock. We formalize and microfound this propagation mechanism and show that it explains both established aggregate moment dynamics and a new size-gradient fact: in a typical quarter, growth rates are substantially more dispersed among smaller firms, while the skewness of growth among larger firms fluctuates much more strongly over the business cycle.

We begin with three facts about quarterly real-sales growth among U.S. firms in Compustat from 1984Q1 through 2024Q4. Mean sales growth falls sharply in recessions, with the largest positive observations occurring mainly during subsequent recoveries.¹ Dispersion is V-shaped around trend growth: it rises in both large contractions and large expansions. Skewness is strongly procyclical: downturns disproportionately generate firms with exceptionally large sales contractions, whereas strong growth disproportionately generates firms with exceptionally large sales increases. Thus the growth distribution becomes left-skewed in downturns and right-skewed when growth is strong.²

The behavior of these moments differs sharply by firm size. Larger firms have lower growth dispersion on average, and their dispersion also changes less over time. Skewness displays the opposite amplitude gradient: its time-series standard deviation rises roughly fourfold from the smallest to the largest size decile. We refer to this joint pattern in the fluctuations of dispersion and skewness as the *opposing size gradients in growth moments*, or, for short, the *size gradient*.

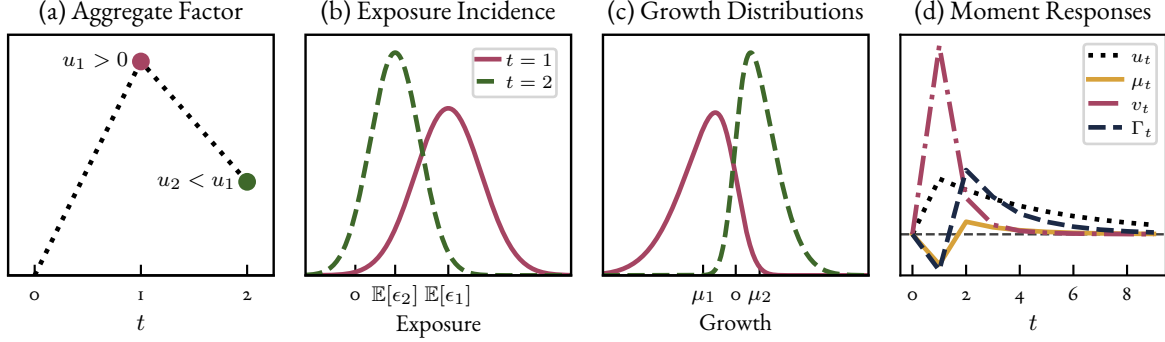
Such distributional fluctuations in firm growth pose a challenge for business-cycle modeling. One natural approach assigns movements in different moments to different shock processes: first-moment shocks for average growth, uncertainty shocks for dispersion, and skewness or tail-risk shocks for asymmetry. Such shocks can be useful, but an explanation of the evidence must account for more than the movements of each moment separately. It must also explain their cyclical comovement and why the variability of dispersion and skewness changes in opposite directions with firm size. These joint requirements motivate our search for a unifying propagation mechanism through which aggregate shocks generate both the cyclical behavior of growth moments and their opposing size gradients.

To build the mechanism, consider a given cross section of firms and suppose there is an aggregate factor that affects all firms' marginal costs, but firms differ in their exposure to it (e.g. Davis et al., 2025). Such a factor could, for example, be the price of a primary input such as oil or energy. When the aggregate factor changes, firms with different exposures, or *loadings*, receive differently sized effective shocks, creating cross-sectional dispersion. If firms' equilibrium mappings from the shock into real revenue—their real-revenue policies—are log-concave, the same cross section of shock incidence maps

¹This asymmetry has been documented at least since Sichel (1993).

²This procyclicality was first emphasized by Salgado et al. (2025). We use left-skewed (right-skewed) synonymously with negatively skewed (positively skewed).

Figure 1: Distributional Shifts in Theory



Note: Panel (a) shows an aggregate cost increase followed by a partial reversal. Panel (b) illustrates how heterogeneous exposure distributes the resulting cost shocks across firms, abstracting from idiosyncratic shocks. Panel (c) shows the corresponding growth distributions: left-skewed on impact and right-skewed during recovery. Panel (d) traces the aggregate factor u_t , mean growth μ_t , variance v_t , and skewness Γ_t as the shock dissipates.

into asymmetric real-revenue growth. On impact of an adverse aggregate shock, adversely exposed firms contract sharply and form a long left tail. Weakly or favorably exposed firms do not generate an equally long right tail because concavity compresses favorable responses. As the shock unwinds, the firms hit hardest on impact exhibit the largest recoveries, producing a right tail and turning skewness positive. Figure 1 illustrates this logic: the aggregate impulse moves the mean, heterogeneous exposure generates dispersion, and local concavity maps dispersed incidence into skewness. The three moments are therefore outcomes of one propagation mechanism rather than reactions to unrelated shocks.

We establish this mechanism formally by characterizing growth moments among firms that share the same policy but differ in their exposure to a common aggregate shock. To second order, growth variance scales with the squared local slope of the real-revenue policy, while growth skewness scales with its normalized local curvature. Under the same aggregate disturbance, steeper policies generate higher and more time-varying cross-sectional growth variance, whereas greater curvature magnitude generates larger fluctuations in cross-sectional skewness. Note that these statements concern time-series moments of cross-sectional moments.³ The result is stated for real-revenue growth rates, our empirical object, and has an analogous output version.

Next, we microfound this concavity in a homothetic monopolistic-competition framework with isoelastic costs. The equilibrium uses aggregate-scale conjectures, an information structure closely related to the *oblivious equilibrium* of Weintraub et al. (2008). Homothetic demand maps the publicly observed aggregate shock into a common scale, and each firm conditions its output choice on this scale and its own cost shock, but not on its rivals' realized idiosyncratic shocks. Firms therefore optimize against perceived inverse-demand functions, which are close to actual inverse demand in equilibrium (see also Negishi, 1961). Actual inverse demand determines the realized prices that clear the product market; these prices generally differ from firms' anticipated prices. The shape of perceived inverse demand determines the shape of firms' perceived marginal-revenue schedules. These schedules, in

³The simplest example of a moment of a moment is the long-run average of a time-varying mean.

turn, determine the shape of firms' equilibrium log-output policies. With a concave policy, cost increases and reductions are passed through asymmetrically to consumers: a 1 percent increase in costs contracts physical output proportionally more than a 1 percent reduction in costs expands it. A strengthened version of Marshall's Second Law (MSLD*) makes these policies concave, while its economic opposite, anti-MSLD*, implies convexity.⁴ This yields a simple comparison: with the same isoelastic cost technology, a price taker has a linear log-output policy, whereas a price-setting firm has a flatter policy that is concave under MSLD*. Interpreting price taking as a benchmark for small firms gives a first account of why larger firms can respond less strongly but more asymmetrically to shocks.

We then extend this comparison to a continuous size gradient among price-setting firms. Differences in productivity place firms at different baseline operating quantities—that is, the output each firm would choose in the absence of shocks. Near a regular finite choke price, a marginal increase in operating quantity reduces the absolute policy slope and makes curvature more negative. The response thus becomes progressively weaker but more asymmetric as firm size increases in this neighborhood. These results hold for both output and real revenue. Thus, size, measured by either output or revenue, becomes associated with less variable growth but larger skewness fluctuations that move in phase with aggregate shocks.

We assess this transmission mechanism through three complementary empirical exercises. First, the component of firm growth transmitted by adverse aggregate innovations should be more dispersed among small firms and more negatively skewed among large firms. We estimate responses to six identified aggregate innovations firm by firm, following a similar approach to Krusell et al. (2026), and use these estimates to construct distributional responses from the bottom up. We then compare their dispersion and skewness across firm-size groups within industries. At the horizon of peak mean contraction, responses are significantly more dispersed among small firms for four shocks and more negatively skewed among large firms for four.

Second, we estimate policy slope and curvature rather than inferring them directly from the raw moment gradients. Our framework jointly recovers relative policy slopes, curvature, and idiosyncratic-shock volatility across size quantiles from the time-series behavior of growth variance and skewness. Its central prediction is borne out in the data: smaller firms have steeper yet less curved policies, notwithstanding their higher idiosyncratic risk. From the smallest to the largest size decile, estimated policy responsiveness roughly halves, while curvature magnitude more than doubles after accounting for differences in idiosyncratic risk.

Third, firms with greater exposure to an aggregate shock should exhibit larger distributional responses. Our COVID event study, standardized to a common industry-size composition, shows a larger variance increase and a more negative skewness response among firms in the outer COVID exposure quintiles (Davis et al., 2025).

Several alternative explanations could generate the raw size gradients without the proposed demand-curvature mechanism. First, idiosyncratic risk may differ systematically across size groups, raising growth variance and mechanically attenuating standardized skewness. Our moment inversion estimates

⁴CES demand is the knife-edge benchmark in which this demand-curvature channel is absent.

this risk jointly with policy slope and curvature. Although idiosyncratic volatility is higher among smaller firms, the opposing slope and curvature gradients survive the resulting correction. Second, shock exposures may vary with size. We show that, in our framework, differences in exposure location and scale are theoretically incompatible with the opposing size gradients when idiosyncratic risk is negligible. An exposure-based explanation therefore requires differences in distributional shape. The COVID and inflation exposure proxies of Davis et al. (2025) do not support the required differences: the relevant exposure moments are nonmonotonic across firm sizes, and we cannot reject equal moments across size groups. Exposure skewness likewise shows no systematic size ordering. Third, the policy differences could reflect technology rather than demand. We reproduce the main moment gradients and recovered policy profiles using both economy-wide size ranks, which permit between-industry composition to matter, and size ranks defined within two-digit NAICS industries, which limit the role of between-industry composition. The results are strikingly similar. A technological explanation would therefore require the elasticity of costs with respect to output, and how that elasticity changes with output, to vary systematically across the size distribution even within industries. Merely assigning firms different exponents in isoelastic cost functions cannot generate the required curvature because such technologies are log-linear. More elaborate production-side mechanisms may generate concave policies (e.g. Miranda-Pinto et al., 2023; Dew-Becker and Vedolin, 2021; Kamepalli et al., 2025), but they would need to produce both a declining absolute policy slope and a rising curvature magnitude within industries.

The paper contributes to the literature in three ways. First, we document a new joint pattern in the cross-sectional moments of firm growth. Procyclical firm-growth skewness is known (Salgado et al., 2025), as is size-dependent cyclicity in firm outcomes (Crouzet and Mehrotra, 2020). Our new empirical fact is the opposing size ordering of dispersion and skewness: larger public firms have lower and more stable growth dispersion, yet substantially greater fluctuations in skewness.

Second, we develop an empirical framework for recovering how aggregate shocks are transmitted through heterogeneous firms. From the joint behavior of growth variance and skewness over time, the framework identifies relative policy slopes and curvature together with idiosyncratic risk. Because the data measure sales rather than physical quantities, we derive this identification directly for real revenue rather than treating deflated sales as a proxy for output.

Third, we provide a demand-side microfoundation for the opposing slope and curvature gradients. We characterize conditions on inverse demand, expressed through MSLD* and a regular finite choke price, under which the absolute policy slope declines and curvature magnitude rises with firm size. This connects the dynamic cross-sectional moments of firm growth to primitive restrictions on demand. To the best of our knowledge, existing models do not jointly explain the aggregate moment dynamics and the two opposing size gradients.

Literature Our paper relates to several strands of the literature on firm growth and aggregate fluctuations. Relative to Salgado et al. (2025), who document procyclical skewness in firm outcomes, the new fact is the size gradient in skewness amplitude and its opposing ordering with dispersion.

Relative to Ilut et al. (2018), who show that concave firm policies can turn symmetric shocks into skewed growth, we derive demand-side policy curvature and study growth as the first difference of a fixed policy. Relative to Davis et al. (2025), who measure heterogeneous firm exposure to aggregate disturbances, we connect exposure to higher growth moments and to the recovery of relative policy objects. Relative to Crouzet and Mehrotra (2020), who study size-dependent cyclicalities, we document the opposing second- and third-moment size gradients. Related work treats skewness as a macroeconomic state variable or forecasting object (Iseringhausen et al., 2026; Dew-Becker, 2024), or studies other sources of aggregate asymmetry (Kamepalli et al., 2025; Dew-Becker and Vedolin, 2021).

Firm heterogeneity and exposure. The empirical facts build on work showing that business cycles and firm growth are distributional. Business cycles are asymmetric (Sichel, 1993; McKay and Reis, 2008; Morley and Piger, 2012; Berger et al., 2020; Dupraz et al., 2025), firm-growth distributions have higher moments that move over the cycle (Higson et al., 2002, 2004), and the firm-growth literature around Gibrat’s Law shows that growth is not simply scale invariant or Gaussian (Sutton, 1997; Bottazzi and Secchi, 2006; Sterk et al., 2021). Firm responses also vary systematically by size and age (Gertler and Gilchrist, 1994; Fort et al., 2013; Crouzet and Mehrotra, 2020). Our empirical implementation is closest to Davis et al. (2025), who measure heterogeneous firm exposure to macro shocks. We use such exposure heterogeneity to connect aggregate shocks to dispersion, complementing uncertainty and endogenous-dispersion accounts of cross-sectional fluctuations (Bloom, 2009; Arellano et al., 2019; Schaal, 2017; Bloom et al., 2018; Kehrig, 2015; Decker et al., 2016; Straub and Ulbricht, 2019).

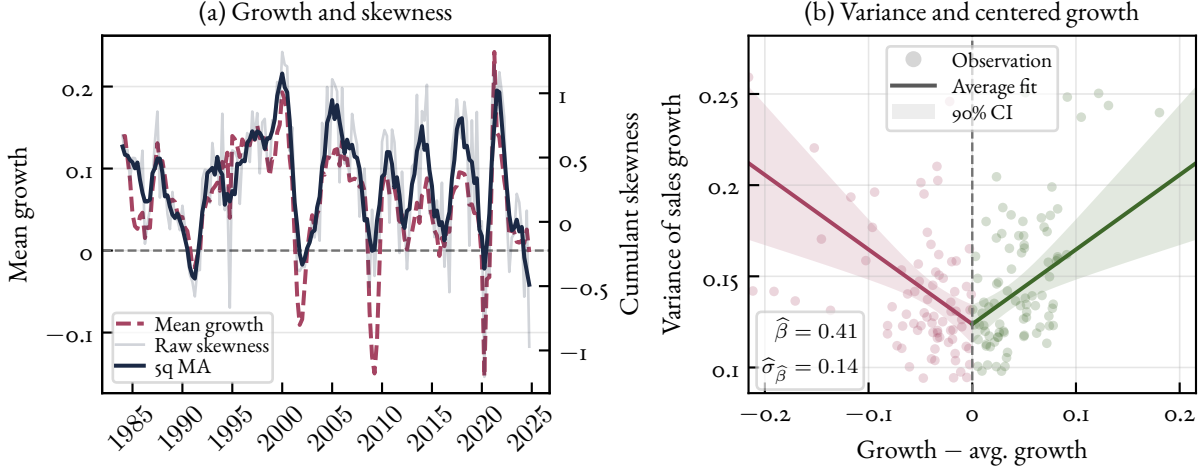
Demand shape and firm policies. The microfoundation draws on non-CES demand and variable-elasticity monopolistic competition. Marshall’s Second Law and the elasticity of marginal revenue determine how cost shocks map into quantities and real revenue (Feenstra, 2003; Parenti et al., 2017; Mrázová and Neary, 2017; Melitz, 2018; Matsuyama and Ushchev, 2022). Empirical product-mix and exporter responses provide related evidence that adjustment varies along the demand curve (Berman et al., 2012; Amiti et al., 2019; Mayer et al., 2021). We add a business-cycle implication: near a regular finite choke price, the shape of inverse demand can make policy slopes fall and curvature magnitudes rise with firm operating size, thereby ordering dispersion and skewness in opposite directions.

Outline Section 2 documents the growth-distribution facts. Sections 3 and 4 develop the moment decomposition and its demand-side microfoundation. Section 5 recovers the relative policy objects and evaluates common-shock propagation and heterogeneous exposure. Section 6 concludes.

2 Stylized Facts

We begin with three facts about the cross section of firm growth. Mean growth falls sharply in recessions; dispersion is low near trend, rises as aggregate growth deviates from trend, and is generally higher for small firms. Skewness is procyclical, with larger fluctuations among large firms.

Figure 2: Growth, Cumulant Skewness, and Dispersion Around Trend



Note: Panel (a) plots mean year-on-year real-sales growth, cumulant skewness, and its centered five-quarter moving average. Panel (b) plots variance against mean growth relative to its time average. The lines show a symmetric V-shaped fit. Shaded regions are 90% Newey–West confidence intervals with four lags.

Data and moments Our baseline sample covers U.S. public firms in Compustat from 1984Q1 through 2024Q4, with the top and bottom 1 percent of firm growth observations trimmed each quarter to limit the influence of outliers. Appendix A provides further details on sample construction.⁵ To study how growth patterns vary with firm size, we measure size throughout the main text by average real quarterly sales over $t - 4, \dots, t - 1$ and compare firms within two-digit NAICS industries. In this section, these comparisons use size deciles formed within each industry every quarter.

Let $R_{it} = p_{it}q_{it}$ be firm i 's nominal sales at time t , P_t the aggregate price level, and $r_{it} = R_{it}/P_t$ real sales. We measure firm growth by the year-on-year change $g_{it} = \log r_{it} - \log r_{i,t-4}$. To track the mean, dispersion, and skewness of firm growth, we compute in each quarter the cross-sectional mean $\bar{g}_t$, variance v_t , and cumulant skewness

$$\Gamma_t := \frac{\mathbb{E}_i[(g_{it} - \bar{g}_t)^3]}{v_t^{3/2}}. \quad (1)$$

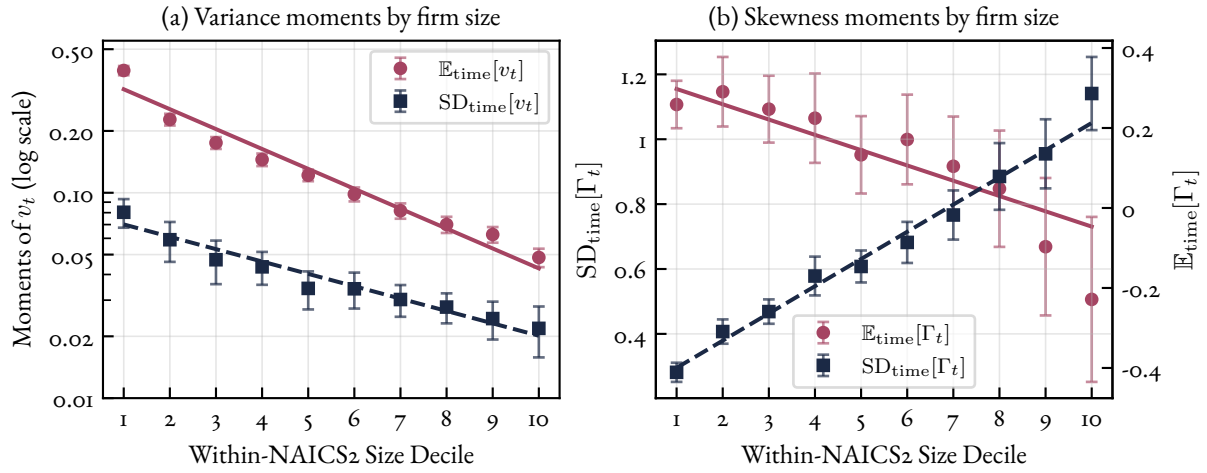
Negative skewness indicates a longer or heavier left tail, and positive skewness the reverse.

Stylized Fact 1: Mean growth follows a ‘plucking’ pattern, normally fluctuating around an expansion trend and occasionally falling sharply around recessions. The dashed red line in panel (a) of Figure 2 shows that mean firm sales growth usually stays near its expansion trend but falls sharply around recessions. These declines are often followed by sharp rebounds, most visibly after the 2009 and 2020 contractions.⁶ This pattern echoes the familiar asymmetry of aggregate fluctuations: contractions tend to be sharp and deep, while expansions are gradual (Sichel, 1993; Morley and Piger,

⁵Table A.1 reports summary statistics and compares these firms with the Quarterly Financial Report (QFR) manufacturing sample of Crouzet and Mehrotra (2020). The comparison makes clear that our sample captures the upper part of the firm-size distribution, within which we study size heterogeneity.

⁶Appendix Figure B.1 shows the mean-growth distribution and the contraction–recovery chronology.

Figure 3: Opposing Size Gradients in Growth Moments



Note: Circles and squares report time-series means and standard deviations of growth variance in Panel (a) and cumulant skewness in Panel (b). Bars are 90% delta-method Newey–West intervals with four lags. Lines show log-linear fits in panel (a) and linear fits in panel (b), weighted by the inverse variance of the estimated moments.

2012; Jensen et al., 2020). It also resembles the plucking dynamics of Dupraz et al. (2025), in which the economy is pulled below trend and subsequently recovers.

Stylized Fact 2: Growth dispersion rises when mean growth moves away from trend. The average level and the time-series variation of dispersion both decline with firm size. Panel (b) of Figure 2 plots the cross-sectional variance of firm growth against deviations of mean growth from its time average. Dispersion is lowest near average growth and rises on either side, tracing a V-shape. We summarize this pattern with $v_t = \alpha + \beta|\bar{g}_t - \bar{g}| + \varepsilon_t$, where $\bar{g}$ is the time average of mean growth. The estimated coefficient is positive and significant. High dispersion is thus not confined to recessions: sharp contractions and sharp rebounds both come with greater differences in firm growth.

Dispersion also exhibits a strong size gradient, both in its average level and in its variation over time. To show this, we compare these moments across the within-industry size deciles.⁷ Panel (a) of Figure 3 shows that both the time-series mean $\mathbb{E}_{\text{time}}[v_{jt}]$ and standard deviation $\text{SD}_{\text{time}}(v_{jt})$ decline monotonically with size, where v_{jt} is the cross-sectional variance of growth in decile j . Larger firms thus have lower dispersion on average, consistent with Crouzet and Mehrotra (2020), and their dispersion varies less over time. So, the size gradient in levels extends to fluctuations in dispersion itself.

Stylized Fact 3: Cumulant skewness is strongly procyclical, and its time-series variability rises steeply with firm size. Firm-growth skewness is strongly procyclical, as emphasized by Salgado et al. (2025). Panel (a) of Figure 2 confirms this pattern for cumulant skewness: the correlation between $\bar{g}_t$ and Γ_t is 0.78. Downturns come with a more left-skewed distribution of firm growth; recoveries with a more right-skewed one.

⁷As mentioned, the within-industry size ranks are our baseline in the main analysis. Appendix Figure B.2 shows that the opposing size gradients are nearly unchanged when firms are instead ranked in the economy-wide size distribution, without using industry information.

The same comovement appears in every size group, with correlations between each decile's skewness and aggregate mean growth ranging from 0.46 to 0.71. What changes with size is the amplitude of skewness fluctuations. Panel (b) of Figure 3 reports the time-series mean and standard deviation of Γ_{jt} in each size decile j . The standard deviation rises monotonically from about 0.28 in the smallest decile to 1.14 in the largest. Average skewness tends to decline with size, though this pattern is less precise.⁸ The key finding is thus an opposing size gradient: larger firms have lower and more stable dispersion, yet larger fluctuations in skewness.⁹ Explaining this pattern jointly with the cyclical comovement of growth moments requires a framework that links aggregate disturbances to differences in firm responses. The next section develops this link through heterogeneous exposure and firm-policy slope and curvature; the demand model then explains how these policy properties vary with firm size.

3 Firm Policies and the Growth Distribution

This section develops a moment decomposition that links firm-level shocks to the cross-sectional variance and skewness of growth via a fixed policy function. Here, we work with the policy in reduced form; later, in Section 4, we derive it from a homothetic demand equilibrium. Note that, even after deflating by the common price level, real sales combine physical output with firm-level relative prices, and thus should not be taken as a proxy for output. We therefore formulate the reduced form both in terms of unobserved output q_{it} and real sales r_{it} to match our empirical measure.

3.1 Reduced-Form Environment

Policy Representation. Firm i is subject to a centered firm-specific shock ϵ_{it} , with cross-sectional mean $\mathbb{E}_i[\epsilon_{it}] = 0 =: \epsilon^*$, standard deviation $\sigma_{\epsilon,t}$ and bounded support, and an aggregate cost shifter ξ_t common to all firms. We write, for physical output and real revenue, respectively,

$$\log q_{it} = a_Q t - b_Q \xi_t + \mathcal{Q}(\epsilon_{it}) \quad \text{and} \quad \log r_{it} = at - b_R \xi_t + \mathcal{R}(\epsilon_{it}) + \zeta_{it}. \quad (2)$$

This reduced form states that log output grows at a deterministic trend $a_Q \geq 0$, responds to the aggregate cost shock with common coefficient $-b_Q$, where $b_Q \geq 0$,¹⁰ and responds to the firm-specific shock through the log-output policy $\mathcal{Q}$.

The real-revenue equation has the same structure, with trend a , aggregate-shock coefficient b_R , and log-revenue policy $\mathcal{R}$, but contains an additional term capturing pricing errors. Firms choose their output exactly, but limited information about their competitors prevents them from perfectly forecasting the price at which it sells. The residual ζ_{it} is the logarithm of the ratio of the realized price

⁸We note a mildly positive average skewness over time. This may reflect secular innovation, entry and exit, or skewed idiosyncratic shocks. For innovation-based sources of right-skewed growth, see, for example, the literature on quality-ladder models or, more recently, Berlingieri et al. (2025). The relevant fact for our mechanism is the cyclical movement of skewness with aggregate growth.

⁹The broad patterns also appear in the Compustat Global sample of non-U.S. firms (Appendix Figures B.3 and B.4).

¹⁰A positive value of ξ_t thus represents a positive cost shock.

to the forecast price. We allow these errors but require them to be common across firms up to second order in the scale of the shock. Formally, there exists a common scalar Z_t such that

$$\zeta_{it} = Z_t + \nu_{it}, \quad (\mathbb{E}_i |\nu_{it}|^3)^{1/3} = O(\sigma_{\epsilon,t}^3). \quad (3)$$

The restriction concerns forecast errors, not prices themselves: firms can charge different prices, but receive approximately the same proportional surprise relative to their forecasts. The common component Z_t shifts every firm's log revenue equally and therefore contributes only to mean growth. Differences in forecast errors across firms enter at third order and therefore do not affect the variance and skewness approximations derived below. Like the policy structure, this restriction will follow from the demand equilibrium we develop in Section 4.1.

Taking first differences gives real-revenue and output growth:

$$g_{it} = a + \Delta \mathcal{R}(\epsilon_{it}) - b_R \Delta \xi_t + \Delta \zeta_{it}, \quad g_{it}^q = a_Q + \Delta \mathcal{Q}(\epsilon_{it}) - b_Q \Delta \xi_t. \quad (4)$$

Real sales growth preserves the fundamental separation of the idiosyncratic and aggregate components as long as the change in the residual is negligible: $-b_R \Delta \xi_t$ shifts the center of the growth distribution, while $\Delta \mathcal{R}(\epsilon_{it})$ determines its shape through shock incidence and policy curvature. The same separation holds exactly for output growth through $\mathcal{Q}$.

Note that our representation differs markedly from Ilut et al. (2018) and Straub and Ulbricht (2019), who write growth as $g_{it} = f(s_{it})$ for a time- t signal s_{it} and a concave map f . In that class of models, higher moments move because the economy shifts the location at which a nonlinear growth-response rule is evaluated. In our formulation, ϵ_{it} never shifts the mean at first order, and the growth policies $\mathcal{R}$ and $\mathcal{Q}$ are always evaluated near the same location, $\epsilon^* = 0$. This separation is useful empirically and theoretically: the slope and curvature of the policies at the single point $\epsilon^* = 0$ characterize the variance and skewness of growth. Without this additive separation of the mean, one would instead need a global characterization of policy slope and curvature, a much harder task. Equation (4) also makes the timing explicit: growth contains the policy difference $\mathcal{R}(\epsilon_{it}) - \mathcal{R}(\epsilon_{i,t-1})$, together with the common change $-b_R(\xi_t - \xi_{t-1})$, rather than applying a nonlinear map only to a time- t signal.

Exposure Structure. Equation (4) suggests two margins of shock exposure: common movements that shift average growth and cross-firm differences that change the shape of the growth distribution. We now give these margins a factor structure. Let there be an aggregate factor u_t , to which each firm i has total exposure $\tilde{\lambda}_i u_t$, where $\tilde{\lambda}_i$ is a firm-specific loading. The average loading across firms i , $\bar{\lambda} := \mathbb{E}_i[\tilde{\lambda}_i] \geq 0$, captures the part of the aggregate factor that moves all firms together. Deviations from that average capture the part that spreads firms out around the cross-sectional mean. We therefore define centered exposure by $\lambda_i := \tilde{\lambda}_i - \bar{\lambda}$, with bounded support, symmetric density Λ , and variance $\sigma_\lambda^2 := \mathbb{V}(\lambda) > 0$. Importantly, since we are agnostic about the origin of the aggregate shock, we are also agnostic about the λ_i 's. One straightforward source of heterogeneous exposure is, for example, a production network in which λ_i captures the importance of a particular intermediate input to firm i

and u_t captures its market price.

In addition to its aggregate exposure, firm i receives a purely idiosyncratic shock $\kappa_{i,t}$. These shocks are iid across firms and time, independent of λ_i , and drawn from a time-invariant symmetric distribution with bounded support, mean zero, and variance σ_κ^2 . The resulting shock, scaling firm i 's marginal cost by $\exp(\tilde{\epsilon}_{i,t})$, is

$$\tilde{\epsilon}_{i,t} = \underbrace{\kappa_{i,t} + \lambda_i u_t}_{\text{centered component}} + \underbrace{\bar{\lambda} u_t}_{\text{common component}}, \quad \bar{\lambda}, u_t \geq 0. \quad (5)$$

The same aggregate factor therefore enters firm growth in two ways. Its average incidence shifts all firms together; differences in exposure move firms relative to one another. In the notation of Equation (2), we set $\xi_t = u_t$, absorb average exposure $\bar{\lambda}$ into the common coefficients b_R and b_Q , and write $\epsilon_{it} = \kappa_{it} + \lambda_i u_t$. We restrict $u_t \geq 0$, so that, together with $\bar{\lambda} \geq 0$, the common cost component is nonnegative. The aggregate factor can push activity below its no-shock benchmark; as it unwinds, activity recovers. This gives the factor the plucking interpretation motivated by Stylized Fact 1.¹¹

Substituting the factor structure into Equation (4) yields growth rates

$$g_{i,t} = a + \Delta \mathcal{R}(\epsilon_{i,t}) - b_R \Delta u_t + \Delta \zeta_{i,t}, \quad g_{i,t}^q = a_Q + \Delta \mathcal{Q}(\epsilon_{i,t}) - b_Q \Delta u_t. \quad (6)$$

Thus, the common term $-b_R \Delta u_t$ moves the center of the real-sales growth distribution, while the policy-difference term governs its shape. When u_t rises, the common term shifts the distribution left, while firms with different λ_i 's move by different amounts across the policy. Whether this spread remains symmetric or becomes skewed depends on the local curvature of $\mathcal{R}$. The same logic applies to output growth through $\mathcal{Q}$. We therefore turn to a measure of policy curvature relative to slope.

Local Curvature Index. To measure local curvature, define for any twice differentiable log policy Y and shock value ϵ :

$$\rho[Y](\epsilon) := \frac{Y''(\epsilon)}{|Y'(\epsilon)|}. \quad (7)$$

Negative values correspond to local concavity, and more strongly negative values mean stronger concavity relative to the policy's slope. At the no-shock benchmark $\epsilon^* = 0$, write $\rho_Y^* := \rho[Y](0)$. For physical output, $Y = \mathcal{Q}$ and we write $\rho_Q^* := \rho[\mathcal{Q}](0)$. For real revenue, $Y = \mathcal{R}$ and we write ρ_R^* . The index summarizes the concavity around the no-shock equilibrium and determines the degree of asymmetry in firm growth responses to an aggregate shock.

¹¹For cross-sectional growth moments at a given date, we allow any nonnegative adapted process for u_t . To study how these moments vary over time, we additionally require stationarity, ergodicity, and the relevant finite moments. One example is the stationary shot-noise process $u_t = \beta u_{t-1} + N_t$, where $N_t \stackrel{\text{iid}}{\sim} \text{Poisson}(\omega)$ and $\beta \in (0, 1)$: innovations raise costs, and their effects decay geometrically.

3.2 Moment Approximation and Stylized Facts

We now use the reduced form to characterize how heterogeneous exposure and policy shape jointly determine the cross-sectional mean, variance, and skewness of firm growth. Proposition 1 derives second-order approximations to the conditional mean and variance, a leading-order approximation to skewness, and their time-series implications. Although physical output is not observed in our data, the same decomposition applies to output growth after replacing $\mathcal{R}$ and $\rho_{\mathcal{R}}^*$ with $\mathcal{Q}$ and $\rho_{\mathcal{Q}}^*$. These approximations are local to the no-shock benchmark $\tilde{\epsilon}_{it} \approx 0$, i.e., they apply when aggregate and idiosyncratic shocks are small, and are therefore less accurate in extreme periods, such as the COVID shock.

Proposition 1 (Approximate Conditional and Time-Series Growth-Rate Moments). *Let $\mathcal{F}_t$ denote the aggregate information available at time t . Let all required moments be finite, let λ be symmetrically distributed, and let the idiosyncratic shocks $\kappa_{i,t}$ follow the time-invariant symmetric distribution specified above. If real sales growth is given by Equation (6) and the price residual satisfies Equation (3), then the following statements hold up to the stated approximation orders in the scale s_t of $\tilde{\epsilon}_{i,t}$.¹²*

i) *The conditional cross-sectional moments are*

$$\mu_t := \mathbb{E}[g_{it} \mid \mathcal{F}_t] = a - b_R \Delta u_t + \frac{\mathcal{R}''(0)}{2} \Delta(u_t^2) \sigma_\lambda^2 + \Delta Z_t + O(s_t^3), \quad (8)$$

$$v_t := \mathbb{V}(g_{it} \mid \mathcal{F}_t) = \mathcal{R}'(0)^2 ((\Delta u_t)^2 \sigma_\lambda^2 + 2\sigma_\kappa^2) + O(s_t^3), \quad (9)$$

$$\Gamma_t := \gamma_1(g_{it} \mid \mathcal{F}_t) = \rho_{\mathcal{R}}^* \frac{3}{2} \frac{4(\Delta u_t)(u_{t-1} + u_t) \sigma_\kappa^2 \sigma_\lambda^2 + \Delta(u_t^2)(\Delta u_t)^2 \mathbb{V}(\lambda^2)}{(2\sigma_\kappa^2 + (\Delta u_t)^2 \sigma_\lambda^2)^{3/2}} + O(s_t^2). \quad (10)$$

ii) *Suppose additionally that the aggregate process $\{u_t\}_t$ is stationary and ergodic. Holding the shock and exposure processes fixed,*

$$\begin{aligned} \mathbb{E}_{\text{time}}[v_t] &\propto \mathcal{R}'(0)^2, & \text{SD}_{\text{time}}(v_t) &\propto \mathcal{R}'(0)^2, \\ \mathbb{E}_{\text{time}}[\Gamma_t] &\propto \rho_{\mathcal{R}}^*, & \text{SD}_{\text{time}}(\Gamma_t) &\propto |\rho_{\mathcal{R}}^*|. \end{aligned} \quad (\text{II})$$

The output-growth case follows by replacing the policy, trend, aggregate coefficient, and curvature index with their $\mathcal{Q}$ -counterparts, and setting the cross-price residual term in Equation (6) to zero.

The proof calculates a second-order Taylor approximation of Equation (6) in u_t and $\kappa_{i,t}$ around $u_t = \kappa_{i,t} = 0$, and evaluates the moment operators at this approximation. We collect this and all other technical proofs in the Appendix.

Proposition 1 links the shock process and policy shape to the three stylized facts. In the remainder of this subsection, we apply these moment approximations in turn to identify the features of aggregate shocks and firm policies that account for each empirical pattern.

¹²Here, $s_t \downarrow 0$ denotes a common local scale such that both $\tilde{\epsilon}_{i,t}$ and $\tilde{\epsilon}_{i,t-1}$ are $O(s_t)$, uniformly across firms. We write $X_t = O(s_t^k)$ when X_t/s_t^k remains bounded as $s_t \downarrow 0$, and $X_t = \Theta(s_t^k)$ when $|X_t|/s_t^k$ is bounded above and away from zero. For random quantities, these bounds apply in the norm relevant to the stated moment.

Mean and Stylized Fact 1 Equation (8) separates average real-sales growth into three effects. First, when u_t rises, the common cost component increases and the term $-b_R \Delta u_t$ lowers average growth. Second, heterogeneous exposure spreads firms out along the policy $\mathcal{R}$. If this policy is concave, firms receiving favorable shocks expand less than firms receiving equally large adverse shocks contract. This Jensen effect further lowers average growth and is captured by the term involving $\mathcal{R}''(0)$. Finally, ΔZ_t captures the change in the common proportional price forecast error. If realized prices exceed forecasts by more than in the previous period, this term raises measured real-sales growth; if they exceed forecasts by less, it lowers growth. In our symmetric industry equilibrium, this price adjustment enters at second order.¹³ Because it is common across firms at that order, it affects mean growth but leaves the variance and skewness approximations unchanged.

Dispersion and Stylized Fact 2 Equation (9) explains the dispersion pattern in Stylized Fact 2. With fixed exposures λ_i , the aggregate contribution to growth variance v_t depends on $(\Delta u_t)^2$: firms contract by different amounts when u_t rises and recover by different amounts when it falls. Dispersion therefore rises during both contractions and recoveries, consistent with the V-shaped pattern around average growth in Figure 2. Idiosyncratic shocks contribute a constant component to growth variance.

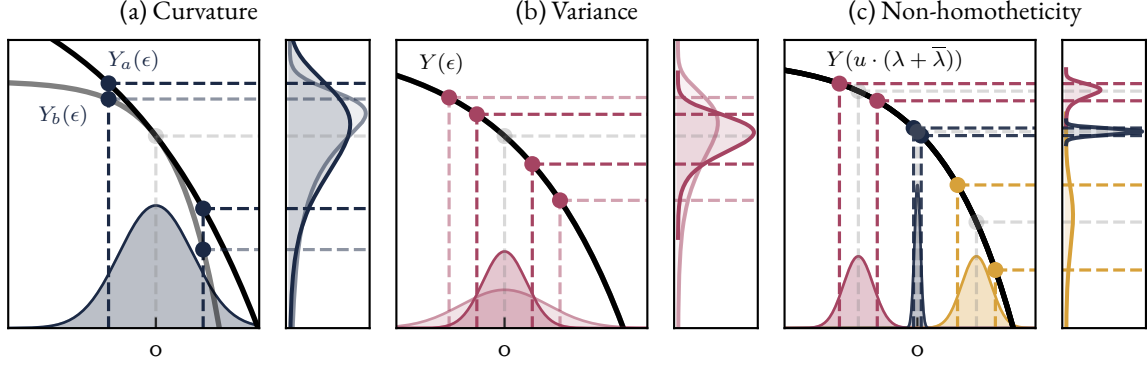
The same equation suggests why dispersion is lower and less variable among larger firms. Holding the shock and exposure processes fixed, both average growth variance and its time-series standard deviation scale with $\mathcal{R}'(0)^2$. Flatter policies among larger firms can therefore account for both size patterns in Stylized Fact 2. Section 3.3 examines possible size differences in exposure variance σ_λ^2 and idiosyncratic variance σ_κ^2 , and Section 5.3 estimates relative policy slopes while allowing idiosyncratic risk to differ across size groups.

Skewness and Stylized Fact 3 Equation (10) shows how policy curvature turns symmetric shock incidence into skewed growth. If the revenue policy were log-linear, its curvature index would be zero and symmetric shocks would generate no skewness through the policy. Concavity breaks this symmetry: firms receiving favorable shocks expand by less than firms receiving equally large adverse shocks contract. Local concavity, $\rho_{\mathcal{R}}^* < 0$, therefore implies a left-skewed distribution of growth upon arrival of an adverse shock. Holding the shock and exposure processes fixed, its magnitude $|\rho_{\mathcal{R}}^*|$ scales the skewness response.

More generally, a negative $\rho_{\mathcal{R}}^*$ accounts for the overall procyclical skewness pattern in Stylized Fact 3. On impact of a contractionary aggregate shock ($\Delta u_t > 0$), mean growth falls and the dispersion of $\lambda_i u_t$ rises. Local concavity maps this wider shock incidence into disproportionately large contractions, so growth skewness turns negative. During recovery, recession-period log revenue is subtracted from current log revenue: the firms that contracted most now recover most. Mean growth rebounds and skewness turns positive. Changes in the aggregate shock are essential here. If the shock remained at its

¹³The characterization of our industry equilibrium in Proposition 3 gives $\Delta Z_t = \chi_h \sigma_\lambda^2 \Delta(u_t^2)$, where $\chi_h = \frac{1}{2} A_0 \mathcal{Q}''(0) + B_0 \mathcal{Q}'(0)^2$. The first component captures the relative-price response to a uniform increase in rivals' quantities and is negative under locally concave output policies and downward-sloping relative demand. The second captures the response to a mean-preserving spread in rivals' quantities and is naturally positive under gross substitutes.

Figure 4: Skewness in Firm Growth: Concavity, Shock Variance, and Non-Homotheticity



Note: Panel (a) maps symmetric shock incidence into skewed outcomes through a concave policy. Greater local concavity under policy b produces more left skewness. Panel (b) holds the policy fixed and increases shock variance, amplifying outcome skewness. Panel (c) illustrates non-homotheticity: aggregate states shift the shock distribution, changing the slopes and curvatures at which firms operate. Colors distinguish states; dashed lines mark the 10th, 50th and 90th percentiles.

high level $u_{t+1} = u_t$ next period, Γ_{t+1} would vanish; a further deepening of the adverse shock would create an even stronger output and revenue asymmetry at date $t + 1$, which would show again as negative Γ_{t+1} .

The local curvature index also helps explain the larger skewness fluctuations among larger firms in Stylized Fact 3. Holding the shock and exposure processes fixed, Equation (10) makes the amplitude of the response proportional to $|\rho_{\mathcal{R}}^*|$. More negative curvature therefore amplifies both sides of the cycle: skewness becomes more negative on impact and more positive during recovery. The observed size gradient thus points to more negative real-revenue curvature among larger firms. This interpretation requires common exposure and idiosyncratic-shock distributions across size groups; the raw gradient alone does not identify curvature. Section 3.3 examines these restrictions, and Section 5.3 estimates relative curvature while allowing idiosyncratic variance to differ across groups.

This reduced-form channel motivates the demand-side microfoundation below. Along a regular inverse-demand curve, we give sufficient conditions under which $|\rho_{\mathcal{R}}|$ rises and $|\mathcal{R}'(0)|$ falls with the firm's operating size. The model therefore jointly rationalizes the comovement of all three moments and the opposing size gradients.

Visualizing the Reduced-Form Channel Since variance transmission is straightforward—a steeper mapping applied to an input distribution spreads the output distribution more—we focus on skewness to illustrate the mechanism. Figure 4 shows the two ingredients of the skewness equation, Equation (10). Panel (a) holds the shock distribution fixed and changes policy curvature: the more concave policy compresses favorable shocks more strongly and produces a more left-skewed outcome distribution. Panel (b) holds the policy fixed and increases the variance of shocks: a wider shock distribution places more mass in the compressed (and stretched-out) regions of the policy, amplifying left skewness. Since Δu_t implies a change in dispersion, taking log first differences essentially subtracts two differently skewed distributions, resulting in a skewness index that is dominated by the more skewed of the two.

Panel (c) illustrates what changes when the aggregate effect cannot be separated additively. The

entire population then evaluates its policies at a different average location after an aggregate shock, so slopes and curvatures at the no-shock benchmark no longer suffice to determine the comovement of dispersion and skewness. Homotheticity in demand supplies the required separation: a proportional change in every firm's cost, or in total market size, changes the common scale while leaving relative quantities and market shares unchanged.

These results show that cyclical skewness does not require a primitive third-moment shock. A mean-shifting aggregate shock is enough: it moves average growth, spreads firms out through heterogeneous exposure, and local concavity maps that spread into skewness. In the baseline, the idiosyncratic-shock distribution is time invariant, so all time variation in the cross-sectional moments is generated by the aggregate factor.

3.3 Threats to Identification

So far, we have interpreted the size gradients through differences in policy slope and curvature, holding shock distributions fixed. However, differences in the distributions of λ and κ across size groups could also contribute to these gradients. We examine their roles in turn, before considering alternative sources of policy curvature.

Exposure Distribution. A natural explanation of size heterogeneity is that aggregate shocks hit smaller firms more unevenly: their exposures have the same distributional shape as those of larger firms, but a greater spread. When idiosyncratic risk is negligible, however, this explanation faces a joint restriction from variance and skewness. Greater exposure dispersion increases both growth dispersion and the amplitude of skewness fluctuations. It therefore cannot explain why smaller firms have more dispersed growth but less variable skewness.

To see this, consider the variance equation, Equation (9), separately for small and large firms, denoted by subscripts S and L . Hold firm policies and the aggregate shock process fixed across groups, and suppose idiosyncratic risk is negligible. Let the only difference be the scale of their centered exposures: the large-firm exposure distribution is obtained by multiplying every loading in the small-firm distribution by $d \in (0, 1)$. The distributions thus have the same shape, but large firms' exposures are less dispersed. Their exposure variance is d^2 times that of small firms, so the equation implies $v_{L,t} \simeq d^2 v_{S,t}$. Large firms consequently have both lower average growth variance and smaller fluctuations in that variance, consistent with the observed dispersion gradient.

The same scaling also predicts smaller skewness fluctuations among large firms. In Equation (10), multiplying exposures by d multiplies $\text{Var}(\lambda^2)$ by d^4 and σ_λ^3 by d^3 . When idiosyncratic risk is negligible, the exposure contribution to skewness is summarized by $C_\lambda := (3/2) \text{Var}(\lambda^2)/\sigma_\lambda^3$, which therefore scales by d . With the same policy curvature in both groups, this implies $\Gamma_{L,t} \simeq d\Gamma_{S,t}$. The data show larger skewness fluctuations among large firms instead. Differences in exposure scale therefore cannot explain the opposing size gradients.

An exposure-based explanation must therefore involve differences in distributional shape, beyond

scale. To reproduce the observed pattern with common policies and negligible idiosyncratic risk, larger firms would need lower exposure variance but a greater skewness-amplification factor C_λ . The COVID and inflation exposure proxies of Davis et al. (2025) allow us to examine these requirements jointly.¹⁴ Appendix B.5 describes the proxies and reports their moments in Figure B.6. For both shocks, the exposure-variance and C_λ profiles are nonmonotonic, with confidence bands that cannot rule out equal moments across size groups. This supplementary evidence does not support the systematic size differences in exposure moments required to explain the opposing growth-moment gradients.

More generally, allowing unrestricted exposure distributions for each size group would give even linear transmission considerable scope to reproduce differences in growth distributions. We restrict exposure to a common, symmetric distribution, to tie our hands and avoid this type of overfitting. While relaxing symmetry would introduce another possible source of growth skewness, an exposure-based explanation would then need to account for its size pattern as well. Under the comparison above, linear transmission would require systematic differences in exposure skewness across size groups. The proxies of Davis et al. (2025) provide a further check: COVID exposures are left-skewed and inflation exposures predominantly right-skewed, but neither exhibits a monotone skewness gradient across firm sizes. Thus, exposure asymmetry may be present, but the systematic size ordering needed for this alternative explanation is not apparent in these proxies.

Idiosyncratic Shocks. The size gradient in growth dispersion need not arise from policy slopes alone. Under the approximation in Proposition 1, the variance within size group j is locally $v_{jt} \simeq \mathcal{R}'_j(0)^2 [2\sigma_{\kappa,j}^2 + \sigma_{\lambda,j}^2(\Delta u_t)^2]$. Thus smaller firms can have more dispersed growth because their policies are steeper, because their aggregate-shock exposures are more dispersed (as discussed), or because their idiosyncratic shocks have larger variance. The first two channels affect how dispersion responds to aggregate shocks. Greater idiosyncratic risk instead raises growth variance by the same amount in every period. It can also attenuate standardized skewness by raising growth variance without a proportionate increase in the third central moment. If $\sigma_{\kappa,j}$ declines with size, large firms can therefore display more variable measured skewness even under identical curvature. Consequently, unconditional size differences in either variance or skewness do not by themselves identify a gradient in policy curvature. At the retained approximation order, however, time-stable $\sigma_{\kappa,j}$ does not affect $\text{SD}_{\text{time}}(v_{jt})$: it shifts the level of variance equally at every date. Size differences in idiosyncratic risk alone therefore cannot explain the declining variability of dispersion in Figure 3.

We account for the effect of idiosyncratic risk when estimating policy curvature. Section 5.3 uses the joint time-series behavior of variance and skewness to estimate relative policy slopes, relative idiosyncratic volatility, and relative curvature magnitudes adjusted for differences in idiosyncratic risk. The factor A_{jt} adjusts skewness for differences in idiosyncratic variance. We find that idiosyncratic volatility is higher among smaller firms, which helps explain their greater average dispersion. After this adjustment, smaller firms still have steeper policies, while relative curvature magnitude rises with firm

¹⁴This evidence is necessarily limited to two shock types during the 2020–2022 period. Moreover, COVID was an unusually large disturbance, so the local moment approximations in Proposition 1 may be less accurate for that episode.

size.

Specification Error. A final threat is specification error: the policy differences that we attribute to demand curvature could instead reflect technological heterogeneity. We control for broad industry composition by using within-industry size rankings throughout the main paper. Each quarter, firms are assigned to sales deciles relative to other firms in the same two-digit NAICS industry, after which equal-ranked firms are pooled across industries. This comparison defines size relative to industry peers, limiting the role of between-industry differences in technology. Appendix B.2 repeats the analysis using economy-wide size rankings, which allow between-industry composition to matter. The moment gradients and recovered policy profiles are similar under the two classifications. A technological explanation must therefore account for the policy gradients within industries. Merely assigning firms different exponents in isoelastic cost functions does not itself introduce cost-side curvature, because marginal cost remains log-linear in output. More elaborate production-side mechanisms may generate concave policies,¹⁵ but they would need to produce both a declining absolute policy slope and rising curvature magnitude across the within-industry size distribution.

4 A Demand-Side Microfoundation

The reduced form identifies policy curvatures, $\rho_{\mathcal{R}}^*$ and $\rho_{\mathcal{Q}}^*$, as the key objects governing skewness. This section provides a demand-side interpretation of both objects. It first derives the policies in Equation (2) in a homothetic industry equilibrium, then decomposes real-revenue curvature into output-policy and relative-price components, and finally asks how their slopes and curvature change with firm size along a common inverse-demand curve.

4.1 Industry Model

We set up an industry equilibrium model of monopolistic competition à la Dixit and Stiglitz (1977) for two purposes. First, it implies the reduced-form structure of log output and real revenue used above. Second, it shows why $\mathcal{Q}$ and $\mathcal{R}$ can be derived from the simpler *partial-equilibrium problem* studied subsequently. Readers interested primarily in the link between demand properties and the shape of policy can therefore proceed directly to Section 4.2; the remainder of this subsection establishes the general-equilibrium foundation for that reduction.

The fundamental assumption is that the underlying preferences are homothetic. Firms own no capital, and therefore solve a static problem. Additionally, firms are information constrained: they know the baseline productivity of each competitor and observe aggregate shocks and their own idiosyncratic shock. They do not observe idiosyncratic shocks—or shock exposures—of their rivals.

¹⁵Supply-side adjustment, capacity constraints, customer capital, and network propagation can generate concave policies.

Households. We consider a representative consumer with a fixed nominal expenditure $E > 0$ spent on a differentiated bundle of goods $\mathbf{q} \equiv (q_i)_{i \in I}$ at prices $\mathbf{p} \equiv (p_i)_{i \in I}$ for some continuum I with unit measure. We use boldface to denote vectors across i . The household has homothetic preferences $U(\mathbf{q}) = F(h(\mathbf{q}))$, where F is strictly increasing and h is strictly increasing, concave, and homogeneous of degree one. On a neighborhood of the baseline equilibrium, h is sufficiently smooth ($h \in C^5$, its derivatives through fifth order are uniformly bounded across varieties, and $h_i > 0$).

The following properties of homothetic demand systems are described in detail in, for example, Parenti et al. (2017): The expenditure function factorizes as $e(\mathbf{p}, \bar{h}) = P(\mathbf{p})\bar{h}$, where $P(\mathbf{p}) \equiv e(\mathbf{p}, 1)$ is the ideal price index (i.e. the minimum expenditure needed to achieve $h = 1$), which is homogeneous of degree one. Marshallian demands are given by $q_i(\mathbf{p}, E) = \frac{E}{P(\mathbf{p})} D_i\left(\frac{\mathbf{p}}{P(\mathbf{p})}\right)$, with $D_i(\mathbf{z}) \equiv \frac{\partial P(\mathbf{z})}{\partial z_i}$. The corresponding inverse-demand system, for any interior output vector $\mathbf{q}$, is

$$p_i(\mathbf{q}, E) \equiv \frac{E}{h(\mathbf{q})} h_i(\mathbf{q}), \quad h_i(\mathbf{q}) \equiv \frac{\partial h(\mathbf{q})}{\partial q_i}.$$

By Euler's theorem, $\int p_i(\mathbf{q}, E) q_i di = \frac{E}{h(\mathbf{q})} \int h_i(\mathbf{q}) q_i di = E$ holds, so total sales indeed equal household expenditure. Moreover, concavity of h implies that $\mathbf{q}$ is utility maximizing at these prices and expenditure E . Thus, $\mathbf{p}(\mathbf{q}, E)$ clears every product market. The associated price index and normalized prices are $P(\mathbf{p}(\mathbf{q}, E)) = \frac{E}{h(\mathbf{q})}$, and $\frac{p_i(\mathbf{q}, E)}{P(\mathbf{p}(\mathbf{q}, E))} = h_i(\mathbf{q})$.

Firms. Since each firm sells exactly one variety i , there is a unit mass of firms. Each firm has a publicly observed baseline cost shifter $\phi_i > 0$, which can be thought of as its productivity inverse, and the entire vector $\boldsymbol{\phi} = (\phi_j)_{j \in I}$ is common knowledge. Firm i 's cost of producing q_i is

$$C_i(q_i; u, \epsilon_i, \phi_i) = \phi_i \exp(\bar{\lambda}u + \epsilon_i) q_i^\eta, \quad \eta > 1,$$

where $\bar{\lambda}u$ is the aggregate effect of an aggregate cost shock u , and ϵ_i is a private, firm-specific shock. This nests the setting of Section 3.1, where $\epsilon_i = \kappa_i + \lambda_i u$. As noted above, $\bar{\lambda}$ and u are publicly observed, but not ϵ_i . At the baseline, all shocks are zero, so firm i 's baseline marginal cost is $\eta \phi_i q_i^{\eta-1}$.

Equilibrium. First consider an equilibrium without firm-specific shocks, where $\epsilon_i = 0$ for all i . Generally, firm i 's nominal revenue from producing q_i given that competitors produce $\bar{\mathbf{q}}_{-i}$ is given by

$$R_i(\mathbf{q}; E) \equiv p_i(\mathbf{q}, E) q_i = E \frac{q_i h_i(\mathbf{q})}{h(\mathbf{q})}$$

and is homogeneous of degree zero in the full output vector. Since expenditure is fixed, $R_i(x; \mathbf{q}_{-i})$ abbreviates $R_i((x, \mathbf{q}_{-i}); E)$. To maximize profits, in an interior equilibrium, each firm i chooses

$$q_i = \arg \max_{x > 0} \{R_i(x; \mathbf{q}_{-i}) - \phi_i e^{\bar{\lambda}u} x^\eta\}. \quad (12)$$

Note that we assume equilibrium uniqueness and interiority throughout. In particular, no firm optimally shuts down production and there is no entry margin. Let $\bar{\mathbf{q}} = \bar{\mathbf{q}}(\boldsymbol{\phi}) \gg 0$ denote the

no-shock benchmark equilibrium, where $u = 0$ and $\epsilon_i = 0$ for all i . Because of the degree-zero homogeneity of R_i , it is easy to see from Equation (12) that $\bar{q}_j(u; \phi) = s(u)\bar{q}_j(\phi)$ with $s(u) \equiv e^{-\bar{\lambda}u/\eta}$ for every j satisfies Equation (12) and thus is precisely the equilibrium for the setting with no firm-level shocks and $u \neq 0$. Hence, the aggregate shock scales all individual responses by a common factor, if there is no heterogeneity in exposure to that shock.

In the case with $\epsilon_j \neq 0$, we assume that firm i optimizes as if $\epsilon_j = 0$ for $j \neq i$. Firm i thus conjectures that the production of other firms is given by $s(u)\bar{\mathbf{q}}_{-i}$; accordingly, it chooses its output to maximize its profits under this conjecture:

$$q_i = \arg \max_{x > 0} \{ R_i(x; s(u)\bar{\mathbf{q}}_{-i}) - \phi_i e^{\bar{\lambda}u + \epsilon_i} x^\eta \}. \quad (13)$$

If firm i sees that the aggregate shock is zero, it solves Equation (13) with $u = 0$. Thus, write $x_i(\epsilon)$ for firm i 's output policy when the aggregate shock is zero (so $\mathbf{x}(0) = \bar{\mathbf{q}}$). The corresponding log-output policy is $\mathcal{Q}_i(\epsilon) = \log x_i(\epsilon)$. The following proposition states that *general* log equilibrium output is $\mathcal{Q}(\epsilon)$ plus a common aggregate level shift. Almost the same result holds for real sales, which are additionally affected by the deviation of a firm's conjectured market clearing price, $p_i(q_i; s(u)\bar{\mathbf{q}}_{-i})$, from the actual market clearing price $p_i(q_i; \mathbf{q}_{-i})$.

Proposition 2 (Industry Equilibrium). *Suppose the firm problems of the equilibrium corresponding to Equation (13) have unique interior solutions, and that households have homothetic preferences. Fix the public environment ϕ and suppress it from notation. Let $s(u) := e^{-\bar{\lambda}u/\eta}$ and $\mathbf{x}(\epsilon) := (x_i(\epsilon_i))_{i \in I}$. Then:*

1. *Firm i 's output choice is $q_i(u, \epsilon_i) = s(u)x_i(\epsilon_i)$. In particular, $x_i(0) = \bar{q}_i$.*

2. *Product markets clear at prices*

$$\tilde{p}_i(u, \epsilon) = s(u)^{-1} \frac{E h_i(\mathbf{x}(\epsilon))}{h(\mathbf{x}(\epsilon))}, \quad P(u, \epsilon) = s(u)^{-1} \frac{E}{h(\mathbf{x}(\epsilon))}.$$

Thus actual prices incorporate both the public baseline cost distribution and the realized cross-section of idiosyncratic shocks, even though each firm ignores competitors' idiosyncratic shocks in its decision problem.

3. *Define the cross-price residual $\zeta_i(\epsilon) := \log h_i(\mathbf{x}(\epsilon)) - \log h_i(x_i(\epsilon_i), \bar{\mathbf{q}}_{-i})$. Then log output and realized log real sales are*

$$\log q_i(u, \epsilon_i) = -\frac{\bar{\lambda}}{\eta}u + \mathcal{Q}_i(\epsilon_i), \quad \log r_i(u, \epsilon) = -\frac{\bar{\lambda}}{\eta}u + \mathcal{R}_i(\epsilon_i) + \zeta_i(\epsilon). \quad (14)$$

Thus, the industry model yields the common aggregate scale in Equation (2), with $b_R = b_Q = \bar{\lambda}/\eta$ when $\xi_t = u_t$. In the exact equilibrium, the cross-price residual ζ_i generally depends on the realized cross-section through $h_i(\mathbf{x}(\epsilon))$.

Proof of Proposition 2. We undo the aggregate shock by setting $q_i = s(u)x_i$. If this substitution turns the conjectured objective into the no-aggregate-shock objective, uniqueness makes its solution the scaled equilibrium; prices and real sales then follow from homogeneity. Under firm i 's conjecture, rivals produce $s(u)\bar{q}_{-i}$. Degree-one homogeneity of h and degree-zero homogeneity of h_i imply

$$R_i(sx; s\bar{q}_{-i}) = E \frac{sx h_i(sx, s\bar{q}_{-i})}{h(sx, s\bar{q}_{-i})} = E \frac{x h_i(x, \bar{q}_{-i})}{h(x, \bar{q}_{-i})} = R_i(x; \bar{q}_{-i}).$$

Moreover, since $s(u) = e^{-\bar{\lambda}u/\eta}$, the cost term satisfies $\phi_i e^{\bar{\lambda}u+\epsilon_i}(sx)^\eta = \phi_i e^{\epsilon_i} x^\eta$. The transformed problem is exactly the zero-aggregate-shock problem defining $x_i(\epsilon_i)$, so uniqueness gives $q_i(u, \epsilon_i) = s(u)x_i(\epsilon_i)$ and $x_i(0) = \bar{q}_i$.

Actual output is $\mathbf{q} = s\mathbf{x}(\epsilon)$. Evaluating inverse demand and the homothetic price index gives

$$\tilde{p}_i = s^{-1} \frac{E h_i(\mathbf{x})}{h(\mathbf{x})}, \quad P = s^{-1} \frac{E}{h(\mathbf{x})}.$$

Euler's theorem yields $\int \tilde{p}_i q_i di = E h(\mathbf{q})^{-1} \int h_i(\mathbf{q}) q_i di = E$, while concavity of h makes $\mathbf{q}$ household-optimal; hence markets clear. Deflated sales are $r_i = \tilde{p}_i q_i / P = s x_i(\epsilon_i) h_i(\mathbf{x})$. Taking logs and adding and subtracting $\log h_i(x_i(\epsilon_i), \bar{q}_{-i})$ gives Equation (14). The output identity follows from $\log q_i = \log s + \log x_i$. ■

The equilibrium concept uses aggregate-scale conjectures: firms condition on the common scale implied by homothetic demand and on their own shock, but not on rivals' idiosyncratic market conditions. It is therefore closely related to the oblivious-equilibrium concept of Weintraub et al. (2008), in which firms condition on their own state and an aggregate representation of the industry state while ignoring competitors' current individual states. In our static environment, the corresponding aggregate state is the publicly observed aggregate shock u , which homotheticity maps into the common scale $s(u)$. Each firm combines this scale with its own shock ϵ_i , without conditioning on ϵ_{-i} . These conjectures are correct in the aggregate-shock-only counterfactual but generally misspecified along the realized path when competitors receive nonzero idiosyncratic shocks or have heterogeneous exposures.

This restriction relieves firms of an informational burden that is difficult to justify empirically (Moll and Ryzhik, 2026). A full-information equilibrium would require firms to observe competitors' shocks, while a rational-expectations formulation would at least require knowledge of their distribution, strategies, and the full demand system. Here, firm i needs to know only the homothetic structure of demand and its own no-shock demand schedule, which it scales by the aggregate shock.¹⁶ In effect, firms solve a partial-equilibrium problem against a perceived own-demand curve, while transactions take place and product markets clear at the realized equilibrium prices. This separation between perceived firm demand and realized market outcomes follows the general-equilibrium approach to perceived demand introduced by Negishi (1961). There is also a formal benchmark for this restricted conditioning. Weintraub et al. (2008) show that, under regularity conditions, oblivious equilibrium closely approximates Markov-perfect equilibrium in a dynamic industry model. Their result does not

¹⁶Note that the model readily extends to a setting with many types of aggregate shocks. We therefore think of industries populated by firms who have strong knowledge about their competitors *not* as industries in which idiosyncratic exposures are public knowledge, but rather as those in which a large number of aggregate factors is known.

directly apply to our static environment, but demonstrates that restricted conditioning can closely approximate full-state equilibrium, lending support to the analogous conditioning rule used here.

The aggregate-scale conjectures also yield the exact common-scale decomposition in Equation (14). Under the shock structure of Equation (5), the firm-specific component is $\epsilon_i = \kappa_i + \lambda_i u$, while the aggregate component remains common. Up to this common component, log output depends only on the firm's own shock through its policy, $\mathcal{Q}$. Real sales admit a similar decomposition but include the additional market-clearing price residual ζ_i . By definition, real sales deflated by the price index are $r_i(u, \epsilon) = \frac{\tilde{p}_i(u, \epsilon) q_i(u, \epsilon_i)}{P(u, \epsilon)}$. Taking logs and using the equilibrium price-index formula in Proposition 2 yields Equation (14):

$$\begin{aligned} \log r_i(u, \epsilon) &= \log q_i(u, \epsilon_i) + \log \frac{\tilde{p}_i(u, \epsilon)}{P(u, \epsilon)} = -\frac{\bar{\lambda}}{\eta} u + \mathcal{Q}_i(\epsilon_i) + \log h_i(\mathbf{x}(\epsilon)) \\ &= -\frac{\bar{\lambda}}{\eta} u + \underbrace{\mathcal{Q}_i(\epsilon_i) + \log h_i(x_i(\epsilon_i), \bar{\mathbf{q}}_{-i})}_{\mathcal{R}_i(\epsilon_i)} + \underbrace{\log h_i(\mathbf{x}(\epsilon)) - \log h_i(x_i(\epsilon_i), \bar{\mathbf{q}}_{-i})}_{\zeta_i(\epsilon)}. \quad (15) \end{aligned}$$

Market clearing introduces firm-specific relative-price adjustments into real revenue. The next proposition shows how symmetry in demand and baseline cost shifters disciplines these adjustments: their second-order component is common across firms, while the remaining firm-specific component is third order. Intuitively, firms' ratios of conjectured to market-clearing prices are equal up to second order despite their different idiosyncratic price choices. Firms therefore make similar forecast errors in logs. The common component may affect the level of log real sales, but not its cross-sectional distribution, while the firm-specific remainder is too high-order to enter the leading moment approximations. Proposition 3 formalizes this result and establishes the residual condition required by Proposition 1.

Proposition 3 (Price Residual Condition in Symmetric Equilibrium). *Consider the equilibrium of Proposition 2 under a symmetric demand system and ϕ . If shocks follow the factor structure of Section 3.1, then the inverse-demand residual ζ_{it} in Equation (14) satisfies the reduced-form residual condition in Equation (3). In particular, it can be written as $\zeta_{it} = Z_t + \nu_{it}$, with $\nu_{it} = O(\sigma_\epsilon^3)$ uniformly in i .*

Proposition 3 shows that the leading variance and skewness implications of firm policies survive market clearing. Symmetry makes the second-order price adjustment Z_t common across firms, so it affects mean growth only. The following remark relaxes this assumption by allowing productivity and demand to differ across types, preserving the moment formulas within each type. Output growth requires no symmetry assumption because quantities are unaffected by the price residual.

Remark 1 (Breaking symmetry.). *Appendix E.1 allows finitely many positive-mass public types $\tau = (\phi_\tau, p_\tau)$, where ϕ_τ is the baseline cost shifter and p_τ is the residual inverse-demand schedule, including its dependence on competitors' quantities. These characteristics can differ across types, while demand remains symmetric within each type.*

Homogeneity then continues to imply $q_{i\tau}(u, \epsilon_{i\tau}) = s(u)x_\tau(\epsilon_{i\tau})$ exactly. Under the maintained shock conditions, the price residual satisfies $\zeta_{i\tau,t} = Z_{\tau,t} + \nu_{i\tau,t}$ with $\|\nu_{\tau,t}\|_{L^3} = O(s_t^3)$, where s_t is the largest type-specific shock scale. Although $Z_{\tau,t}$ can differ across types, within each type it shifts mean growth

only; the leading variance and skewness formulas are unchanged. Thus the key cross-sectional results hold type by type. With a common inverse-demand schedule, size groups can be defined by the baseline cost parameter ϕ_τ , or equivalently by productivity.

Crucially, the model allows us to study higher-order moments of growth through a partial-equilibrium problem. Since the log-output policy $\mathcal{Q}$ is independent of aggregate shocks and rivals' realized output choices, one can derive it by setting $u = 0$ and solving Equation (13). This is equivalent to solving a partial-equilibrium problem with residual inverse demand $p_i(q_i) \propto h_i(q_i; \bar{\mathbf{q}}_{-i})$. It is therefore perfectly valid to study $\mathcal{Q}$ and $\mathcal{R}$ through properties of residual inverse demand p_i only. In this partial equilibrium, the output and revenue policies are connected through

$$\mathcal{R}_i(\epsilon) = \mathcal{Q}_i(\epsilon) + \log p_i(e^{\mathcal{Q}_i(\epsilon)}), \quad (16)$$

where the cross-price residual ζ_i is omitted on the grounds established by Proposition 3. We study this partial-equilibrium problem in the next subsection.

4.2 Firm Problem

Fix a firm from the industry model and suppress firm-specific notation. The firm problem has normalized convex cost function $C(q; \epsilon) = \phi e^\epsilon q^\eta$, where $\eta > 1$, q denotes output, and $\phi > 0$ is the constant cost shifter inherited from the firm-specific environment. The stochastic cost shifter $\exp \epsilon$ is the firm-specific shock, corresponding to ϵ_i above and nesting the factor structure described earlier. We assume that ϵ is drawn from a symmetric distribution with finite variance and is observed by the firm at the time of production. We deliberately keep the cost side iso-elastic so that any local curvature in the output policy comes from demand-side adjustment rather than mechanical cost-side nonlinearities.

The price-setting problem. The firm faces inverse demand $p(q)$ and solves

$$\max_{q \geq 0} qp(q) - C(q; \epsilon). \quad (17)$$

Assumption 1 guarantees that the problem is well behaved and the optimum is unique.

Assumption 1 (Regularity Conditions). *The inverse demand function p satisfies:*

- i) $p : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $p \in \mathcal{C}^3(\mathbb{R}_+)$, and $p'(q) < 0$ on the relevant operating domain,
- ii) marginal revenue $R'(q) := p(q) + qp'(q)$ is strictly decreasing on the relevant operating domain and positive at some finite quantity level.

The condition on marginal revenue is implied by log-concavity of inverse demand. If p is log-concave and downward sloping, then marginal revenue is strictly decreasing wherever it is positive. It is not necessary: CES inverse demand satisfies the condition as well, although it is not log-concave. Under these assumptions, the optimality condition then takes a Lerner-style form. For any differentiable function f with nonzero sign, define its elasticity by $\mathcal{E}f(q) := \frac{qf'(q)}{f(q)}$. Hats denote logarithms. Then, Lemma 1 characterizes the solution to Equation (17).

Lemma 1 (Solution of Firm Problem). *The firm's solution is unique, interior, and implicitly given by*

$$\eta\phi e^\epsilon q^{\eta-1} = p(q)(1 + \mathcal{E}p(q)) =: R'(q), \quad (18)$$

$$\iff \epsilon = \log R'(e^{\hat{q}}) - \log \eta - (\eta - 1)\hat{q} - \log \phi =: H(\hat{q}), \quad \hat{q} := \log q. \quad (19)$$

Optimal log output, $\mathcal{Q}(\epsilon)$, is decreasing in ϵ . On the relevant domain, inverse-demand elasticity satisfies $\mathcal{E}p(q) \in (-1, 0)$ and the markup is $\mu(q) = (1 + \mathcal{E}p(q))^{-1}$.

Lemma 1 gives us a convenient way to study the curvature of the output policy. The first-order condition defines $H(\hat{q})$, which is the inverse of $\hat{q} = \mathcal{Q}(\epsilon)$. We can therefore study the curvature of $\mathcal{Q}$ by studying the curvature of H through the relationship (compare Lemma 2 in the Appendix)

$$\rho_{\mathcal{Q}}^* = \frac{\mathcal{Q}''(\epsilon^*)}{|\mathcal{Q}'(\epsilon^*)|} = \frac{H''(\hat{q}^*)}{H'(\hat{q}^*)^2}.$$

If H is affine, then $\mathcal{Q}$ is affine and symmetric shocks do not generate skewness. Demand matters only insofar as the marginal-revenue schedule bends this affine case.

The price-taking benchmark. The price-taking case is exactly the affine case. A price-taking firm with $\eta > 1$ chooses output taking the market price $\bar{p}$ as given.¹⁷ Its first-order condition is $\epsilon = \log \bar{p} - \log \eta - (\eta - 1)\hat{q}_{\text{pt}} - \log \phi$. The right-hand side is affine in log output, so the inverse policy $\mathcal{Q}_{\text{pt}}(\epsilon)$ is affine in the shock. With no markup-adjustment margin, its benchmark curvature is $\rho_{\text{pt}}^* := \rho[\mathcal{Q}_{\text{pt}}](\epsilon^*) = 0$.

4.3 Growth Moments

Section 3.1 showed that the cross-sectional variance of growth scales with the squared local policy slope, whereas skewness scales with normalized policy curvature. We now derive these two policy objects from demand primitives. We begin with physical output and subsequently turn to real revenue.

Demand-shape objects. From the first-order condition, we see that marginal revenue defines the shape of H , and hence of the policy $\mathcal{Q}$. To characterize the geometry of marginal revenue, define the elasticity and superelasticity of inverse demand together with the elasticity of marginal revenue:

$$\varepsilon(q) := \mathcal{E}p(q) < 0, \quad \psi(q) := \mathcal{E}\varepsilon(q) = \frac{q\varepsilon'(q)}{\varepsilon(q)}, \quad m(q) := \mathcal{E}R'(q) = \frac{qR''(q)}{R'(q)}. \quad (20)$$

Here, ψ is the *superelasticity* of inverse demand (Melitz, 2018). Recall that $\epsilon^* = 0$, and let $q^* := \exp(\mathcal{Q}(\epsilon^*))$ denote the corresponding reference quantity. We write $\varepsilon^* := \varepsilon(q^*)$, $\psi^* := \psi(q^*)$, and $m^* := m(q^*)$. The restrictions below use these objects at the reference quantity to link demand primitives directly to policy slope and curvature:

Definition 1 (Marshallian Second-Law Properties).

¹⁷Strict $\eta > 1$ is needed here to avoid degenerate zero-or-infinity solutions.

- i) Marshall's Second Law of Demand (MSLD) holds at q^* if the absolute elasticity of inverse demand is locally increasing: $-\varepsilon'(q^*) > 0$.
- ii) Marshall's Strong Second Law of Demand (MSLD*) holds at q^* if MSLD holds and the absolute elasticity of marginal revenue is locally increasing: $-m'(q^*) > 0$.
- iii) Anti-MSLD holds at q^* if the absolute elasticity of inverse demand locally decreases, $-\varepsilon'(q^*) < 0$. Anti-MSLD* holds if, additionally, the absolute elasticity of marginal revenue locally decreases, $-m'(q^*) < 0$.

MSLD says that consumers become more sensitive to percentage price changes as the price rises along a demand curve. As a firm expands output and lowers its price, demand therefore becomes less elastic, allowing a higher markup. The firm's first-order condition gives the Lerner index $-\varepsilon(q)$ and markup $\mu(q) = [1 + \varepsilon(q)]^{-1}$, both increasing with quantity under MSLD. Lower-cost firms operating further down a common demand curve thus charge higher markups. Cost reductions are partly absorbed in higher markups, while cost increases compress them, dampening pass-through into prices (Mrázová and Neary, 2017; Melitz, 2018).

MSLD* extends the formalization of this same economic idea to the elasticity of marginal revenue. The two restrictions are closely linked: if the absolute elasticity of marginal revenue increases globally, then MSLD follows as well (Mayer et al., 2021; Melitz, 2018). The anti-laws describe the economic opposite: consumers become less sensitive to percentage price changes at higher prices, with anti-MSLD* also reversing the marginal-revenue condition. The literature generally treats MSLD and MSLD* as the empirically relevant cases, consistent with evidence on incomplete cost pass-through, higher markups among more productive firms, and sales reallocations toward better-performing products (Melitz, 2018; Mayer et al., 2021). We focus on these reinforcing patterns, setting aside locally conflicting signs that capture neither the law nor its economic opposite.

Implications for the variance and skewness of output growth. The firm's first-order condition maps its cost shock into optimal output through the marginal-revenue schedule. The local slope of this policy measures the strength of the output response, while the change in this slope determines whether favorable and adverse shocks have asymmetric effects. The following proposition expresses both objects in terms of the elasticity of marginal revenue.

Proposition 4 (Output policy on a given inverse-demand curve). *Define the marginal cost–revenue elasticity gap as $D(q) := \eta - 1 - m(q)$, the difference between the elasticities of marginal cost and marginal revenue. Under Assumption 1, $D(q^*) > 0$ and*

$$\mathcal{Q}'(0) = -\frac{1}{D(q^*)}, \quad \rho_{\mathcal{Q}}^* = \frac{q^* m'(q^*)}{D(q^*)^2}. \quad (21)$$

Consequently, MSLD implies a locally concave output policy, $\rho_{\mathcal{Q}}^* < 0$, whereas anti-MSLD* implies a locally convex output policy, $\rho_{\mathcal{Q}}^* > 0$.*

Proof of Proposition 4. We apply the inverse-function identities stated in Lemma 2 to the marginal-revenue elasticity. Differentiating H at $\hat{q}^*$ gives $H'(\hat{q}^*) = m(q^*) - (\eta - 1) = -D(q^*)$ and $H''(\hat{q}^*) = q^*m'(q^*)$. Since $m(q^*) < 0$ and $\eta > 1$, $D(q^*) > 0$, and therefore $\mathcal{Q}'(0) = 1/H'(\hat{q}^*) = -1/D(q^*)$. Moreover, $\rho_{\mathcal{Q}}^* = H''(\hat{q}^*)/H'(\hat{q}^*)^2 = q^*m'(q^*)/D(q^*)^2$. Because $q^* > 0$ and the denominator is positive, output-policy curvature has the sign of $m'(q^*)$. MSLD* therefore implies $\rho_{\mathcal{Q}}^* < 0$, while anti-MSLD* implies $\rho_{\mathcal{Q}}^* > 0$. ■

The policy slope is negative because an increase in costs reduces optimal output. Its magnitude, $1/D(q^*)$, measures the strength of this response: a larger marginal cost–revenue elasticity gap implies a flatter output policy. Policy curvature describes how the response changes with the shock. Under MSLD*, a favorable cost shock moves the firm to a higher-quantity region in which marginal revenue is more elastic. The firm then absorbs part of the cost reduction through a higher markup rather than expanding output proportionally. An adverse cost shock moves the firm toward a lower-quantity region in which marginal revenue is less elastic; as a result, the output policy is steeper in the cost shock. The resulting output policy is locally concave.

These properties map directly into the output-growth moments derived in Proposition 1: the squared magnitude of $\mathcal{Q}'(0)$ scales the variance of output growth, while $\rho_{\mathcal{Q}}^*$ governs its skewness response. Since MSLD* makes $\rho_{\mathcal{Q}}^*$ negative, the discussion in Section 3.2 identifies it as the demand property that rationalizes the procyclical movements in skewness documented in Stylized Fact 3.

We now compare a monopolistic (“large”) firm with a price-taking (“small”) firm within the same framework. This comparison provides an initial benchmark for mapping the size gradient into the model.

Corollary 1 (Price-taking benchmark). *The price-taking output policy satisfies $\mathcal{Q}'_{\text{pt}}(0) = -1/(\eta - 1)$ and $\rho_{\text{pt}, \mathcal{Q}}^* = 0$. A price-setting firm satisfies $|\mathcal{Q}'(0)| < |\mathcal{Q}'_{\text{pt}}(0)|$ and, under MSLD*, $\rho_{\mathcal{Q}}^* < \rho_{\text{pt}, \mathcal{Q}}^* = 0$.*

Proof of Corollary 1. Solving the price-taking policy explicitly gives $\mathcal{Q}_{\text{pt}}(\epsilon) = [\log \bar{p} - \log \eta - \log \phi - \epsilon]/(\eta - 1)$, hence $\mathcal{Q}'_{\text{pt}}(0) = -1/(\eta - 1)$ and $\rho_{\text{pt}, \mathcal{Q}}^* = 0$. For a price setter, $m(q^*) < 0$ implies $D(q^*) > \eta - 1$, so Proposition 4 gives $|\mathcal{Q}'(0)| = 1/D(q^*) < 1/(\eta - 1) = |\mathcal{Q}'_{\text{pt}}(0)|$. The same proposition gives $\rho_{\mathcal{Q}}^* = q^*m'(q^*)/D(q^*)^2 < 0$ under MSLD*. ■

The price taker’s first-order condition is log-linear, so its output policy is affine and its skewness response is acyclical. Introducing price setting flattens the output policy and, under MSLD*, introduces negative curvature. It consequently dampens the variance response of output growth while generating an asymmetric response to symmetric shocks and a procyclical skewness pattern over the cycle. This simple comparison suggests that we can generalize the size gradient more formally by considering firms that produce at different operating quantities. This is the task of the next subsection. Before doing so, however, we establish that our conclusions about output growth also apply to revenue.

Log real revenue. The empirical object is real-sales growth rather than physical-output growth. We therefore apply the same argument to the real-revenue policy $\mathcal{R}$, which combines quantity adjustment

with the relative-price adjustment induced by inverse demand. The following proposition characterizes both its slope and its curvature.

Proposition 5 (Real-revenue-policy slope and curvature). *Suppose $p \in C^2$, $p'(q) < 0$, $\varepsilon^* \in (-1, 0)$, and $\mathcal{Q}'(0) \neq 0$. Then the moment approximation in Proposition 1 applies to real-sales growth with policy slope and curvature index*

$$\mathcal{R}'(0) = (1 + \varepsilon^*)\mathcal{Q}'(0) = -\frac{1 + \varepsilon^*}{D(q^*)}, \quad (22)$$

$$\rho_{\mathcal{R}}^* = \rho_{\mathcal{Q}}^* + \frac{\varepsilon^* \psi^*}{1 + \varepsilon^*} |\mathcal{Q}'(0)|. \quad (23)$$

Because $1 + \varepsilon^* \in (0, 1)$, relative-price adjustment makes the real-revenue policy flatter than the output policy. Under MS�D, $\psi^* > 0$, so the relative-price contribution to curvature is negative. Under MS�D*, both curvature contributions are negative, implying $\rho_{\mathcal{R}}^* < 0$. Under anti-MS�D*, both contributions are positive, implying $\rho_{\mathcal{R}}^* > 0$. The price taker satisfies $\mathcal{R}'_{\text{pt}}(0) = \mathcal{Q}'_{\text{pt}}(0)$ and $\rho_{\text{pt}, \mathcal{R}}^* = 0$.

Proof of Proposition 5. Differentiate log real revenue as a composition of inverse demand and the output policy, then sign its two curvature components. Write $q(\epsilon) = e^{\mathcal{Q}(\epsilon)}$. Since $\mathcal{R}(\epsilon) = \mathcal{Q}(\epsilon) + \log p(q(\epsilon))$, the chain rule gives

$$\mathcal{R}'(\epsilon) = [1 + \varepsilon(q)]\mathcal{Q}'(\epsilon), \quad \mathcal{R}''(\epsilon) = [1 + \varepsilon(q)]\mathcal{Q}''(\epsilon) + \varepsilon(q)\psi(q)[\mathcal{Q}'(\epsilon)]^2,$$

where $d\varepsilon(q(\epsilon))/d\epsilon = \varepsilon(q)\psi(q)\mathcal{Q}'(\epsilon)$. Because $1 + \varepsilon^* > 0$, dividing the second identity at zero by $|\mathcal{R}'(0)| = (1 + \varepsilon^*)|\mathcal{Q}'(0)|$ proves Equation (23).

Under MS�D*, $\varepsilon'(q^*) < 0$ and $m'(q^*) < 0$. Since $\varepsilon^* < 0$, the first inequality gives $\psi^* > 0$, making the relative-price contribution in Equation (23) negative. The second inequality gives $\rho_{\mathcal{Q}}^* < 0$ by Proposition 4, so $\rho_{\mathcal{R}}^* < 0$. Under anti-MS�D*, both derivative inequalities reverse: $\psi^* < 0$, the relative-price contribution is positive, and $\rho_{\mathcal{Q}}^* > 0$. Hence $\rho_{\mathcal{R}}^* > 0$. Finally, a price taker has locally constant p , hence $\varepsilon = 0$ and $\mathcal{R} = \mathcal{Q} + \text{constant}$, giving zero curvature. ■

The economics are straightforward. An output adjustment along a downward-sloping demand curve induces an offsetting price movement. This reduces the absolute slope of real revenue and therefore dampens its variance response relative to physical output. Curvature contains two components: curvature of the output policy and curvature introduced directly by relative-price adjustment. Under MS�D, demand becomes less elastic as the firm expands, so price declines offset an increasing share of quantity gains in revenue terms. This adds negative curvature to the real-revenue policy.

Consequently, the same moment interpretation applies to real-sales growth. Its variance scales with $\mathcal{R}'(0)^2$, while its skewness response scales with $\rho_{\mathcal{R}}^*$. At the price-taking benchmark, the relative-price channel disappears, both output and real-revenue policies are affine, and their skewness responses are acyclical. Under MS�D*, price setting instead produces a procyclical real-sales skewness response.

Finally, to illustrate how local concavity translates into skewness impulse-response dynamics, suppose the assumptions of Proposition 1 hold, there are no idiosyncratic shocks, and the aggregate factor follows the mean-reverting path $(u_0, u_1, u_2, \dots) = x \cdot (0, \beta^0, \beta^1, \beta^2, \dots)$, with $x > 0$ and

$\beta \in (0, 1)$. The reduced-form moment approximation yields the following result for both policy objects.

Corollary 2 (Skewness impact and rebound). *Let $Y \in \{\mathcal{Q}, \mathcal{R}\}$, with corresponding growth rate g_t^Y from Equation (6). Let $u_0 = 0$ and $u_t \propto \beta^{t-1}$ for $t > 0$, and let $\sigma_\kappa^2 = 0$. If $\rho_Y^* < 0$, then moment-based skewness is negative on impact and positive during the recovery, with monotone decay:*

$$\gamma_1(g_1^Y) < 0 < \gamma_1(g_t^Y) \quad \forall t > 1, \quad |\gamma_1(g_{t+1}^Y)| = \beta |\gamma_1(g_t^Y)| \quad \forall t \geq 2. \quad (24)$$

Thus, local concavity of $\mathcal{Q}$ gives the impact–rebound prediction for output growth, and local concavity of $\mathcal{R}$ gives the same prediction for real-sales growth. The amplitude of each path is proportional to the corresponding $|\rho_Y^*|$. Under the no-idiosyncratic-shock path used here, the approximation is also homogeneous of degree one in the aggregate shock amplitude x : multiplying x by $c > 0$ multiplies the entire skewness path by c . Thus $|\rho_Y^*|$ ranks responses across policies, while x determines their common scale.

The corollary provides a clean example of how the skewness of growth rates can exhibit a procyclical impulse-response pattern. Unlike Γ_t , however, the cross-sectional variance converges to a positive level proportional to σ_κ^2 when idiosyncratic shocks are present. This unknown long-run level complicates the applicability of impulse-response functions for the second moment in practice.

4.4 The Size Gradient

The preceding subsection takes a firm’s operating point $q = \exp(\mathcal{Q}(0))$, its output at $\epsilon = 0$, as given. We now compare firms that face the same inverse-demand function but operate at different baseline quantities because their baseline productivity differs.

A firm with operating point q is associated with the baseline cost shifter $\phi_q := R'(q)/(\eta q^{\eta-1})$, or, equivalently, with productivity $1/\phi_q$. By construction, q solves this firm’s problem at $\epsilon = 0$. Let $\mathcal{Q}_q$ and $\mathcal{R}_q$ denote the log-output and log-real-revenue policies of such a firm, and define

$$\beta_Q(q) := \mathcal{Q}'_q(0), \quad \rho_Q(q) := \frac{\mathcal{Q}''_q(0)}{|\mathcal{Q}'_q(0)|}, \quad \beta_R(q) := \mathcal{R}'_q(0), \quad \rho_R(q) := \frac{\mathcal{R}''_q(0)}{|\mathcal{R}'_q(0)|}.$$

Thus, $\beta_Q(q)$ and $\beta_R(q)$ are the local policy slopes of a firm whose shock-free operating quantity is q . Their absolute values measure the sensitivity of its output and real revenue to a marginal cost shock. The normalized curvatures $\rho_Q(q)$ and $\rho_R(q)$ measure the asymmetry of these responses and determine their skewness implications through Proposition 1. Variation in these coefficients with q can therefore be interpreted as a size gradient.

We next establish that, near a finite choke price, these coefficients exhibit the size dependency required to match the opposing size gradients in variance and skewness: their absolute slopes decline with operating quantity, while their curvature becomes more negative. Suppose inverse demand has a finite positive choke price, $p(0) = \bar{p} \in (0, \infty)$, and leaves it with a nondegenerate downward slope,

$p'(0) \in (-\infty, 0)$. Proposition 6 shows that these primitive boundary conditions are sufficient for the required output-policy gradients.

Proposition 6 (Output-policy gradients near a finite choke price). *Suppose $p \in C^4([0, \bar{q}])$ for some $\bar{q} > 0$, $p(0) = \bar{p} \in (0, \infty)$, and $p'(0) \in (-\infty, 0)$. Then, as $q \downarrow 0$,*

$$\beta_Q(q) = -\frac{1}{\eta - 1} + C_Q q + O(q^2), \quad \rho_Q(q) = -C_Q q + O(q^2), \quad C_Q := \frac{-2p'(0)}{\bar{p}(\eta - 1)^2} > 0. \quad (25)$$

Consequently, for all sufficiently small $q > 0$,

$$\beta'_Q(q) > 0, \quad \rho'_Q(q) < 0, \quad \frac{d|\beta_Q(q)|}{dq} < 0, \quad \frac{d|\rho_Q(q)|}{dq} > 0.$$

Proof of Proposition 6. At the boundary, $R'(0) = \bar{p}$. Differentiating $\varepsilon(q) = qp'(q)/p(q)$ and $m(q) = qR''(q)/R'(q)$ and taking right limits gives

$$2\varepsilon'_+(0) = m'_+(0) = \frac{R'_+(0)}{R'(0)} = \frac{2p'_+(0)}{\bar{p}} =: -\iota_{R'} < 0. \quad (26)$$

Hence $m(q) = -\iota_{R'} q + O(q^2)$ and $m'(q) = -\iota_{R'} + O(q)$. Set $A := \eta - 1 > 0$. The marginal cost–revenue elasticity gap is $D(q) = A + \iota_{R'} q + O(q^2)$. Proposition 4 gives $\beta_Q = -1/D$ and $\rho_Q = qm'/D^2$. Using

$$\frac{1}{D(q)} = \frac{1}{A} - \frac{\iota_{R'}}{A^2} q + O(q^2),$$

we obtain $\beta_Q(q) = -A^{-1} + \iota_{R'} A^{-2} q + O(q^2)$ and $\rho_Q(q) = -\iota_{R'} A^{-2} q + O(q^2)$, proving the expansions with $C_Q = \iota_{R'}/A^2$. Differentiability of the remainders under $p \in C^4$ gives $\beta'_Q(q) = C_Q + O(q) > 0$ and $\rho'_Q(q) = -C_Q + O(q) < 0$ for sufficiently small $q > 0$. In that neighborhood both β_Q and ρ_Q are negative, so $d|\beta_Q|/dq = -\beta'_Q < 0$ and $d|\rho_Q|/dq = -\rho'_Q > 0$. ■

Under the boundary condition $p'_+(0) \in (-\infty, 0)$, firms producing near the choke price can make accurate demand and revenue calculations using the local slope of inverse demand. Let $\delta > 0$ denote a small price concession below $\bar{p}$. Locally, $\bar{p} - p(q) \approx -p'_+(0)q$. Setting $p = \bar{p} - \delta$ therefore implies $q \approx \delta/[-p'_+(0)]$ and $pq \approx \bar{p}\delta/[-p'_+(0)]$. Thus, output and revenue initially increase proportionally as the price falls below the choke price. In contrast, if $p'_+(0) \in \{0, -\infty\}$, these linear extrapolations are not informative.

Equation (26) shows that $p'_+(0) < 0$ is equivalent to the two inequalities defining MSLD*, evaluated through right-hand derivatives at the boundary. By continuity, MSLD* also holds at sufficiently small positive operating quantities. The boundary expansions therefore establish a comparison between firms facing the same inverse-demand curve but differing in baseline costs: near the choke price, lower-cost firms operate at larger quantities and have flatter, more concave output policies.

Proposition 6 concerns physical output. To connect the model to the object observed in the data, we next establish that the same local size gradients hold for real revenue. Because marginal revenue is positive near the choke price, total revenue is locally increasing in output. Firms can therefore be ranked by real revenue as well as by physical quantity.

Proposition 7 (Real-revenue-policy gradients near a finite choke price). *Under the hypotheses of Proposition 6, as $q \downarrow 0$,*

$$\beta_R(q) = \frac{-1}{\eta - 1} + C_R q + O(q^2), \quad \rho_R(q) = -C_R q + O(q^2), \quad C_R := \frac{-p'(0)}{\bar{p}} \frac{\eta + 1}{(\eta - 1)^2} > 0. \quad (27)$$

Consequently, for all sufficiently small $q > 0$,

$$\beta'_R(q) > 0, \quad \rho'_R(q) < 0, \quad \frac{d|\beta_R(q)|}{dq} < 0, \quad \frac{d|\rho_R(q)|}{dq} > 0.$$

Because $d[p(q)q]/dq = R'(q) > 0$, these are gradients in revenue size and in physical quantity, too.

Propositions 6 and 7 establish the same local size dependency for output and real revenue. Proposition 1 makes growth variance proportional to the squared policy slope, so the decline in $|\beta_R(q)|$ reduces both the aggregate-shock contribution to dispersion and its time variation. The same approximation makes the skewness response proportional to $\rho_R(q)$, so increasingly negative curvature amplifies its cyclical movement. The model can therefore generate the opposing size gradients documented in the data, at least locally: larger firms have less variable growth but more strongly procyclical skewness.

Propositions 6 and 7 impose a finite, nonzero slope at the choke price because it provides a particularly transparent sufficient condition. This condition is stronger than necessary. We can also allow inverse demand to be flat or infinitely steep at the choke price, provided that it approaches the choke price locally like a power function, $p(q) = \bar{p} - cq^\gamma + o(q^\gamma)$, with $c, \gamma > 0$. This is the local form of the generalized-linear inverse-demand family $p(q) = a - bq^\delta$ introduced by Bulow and Pfleiderer (1983). The condition is mild and easy to satisfy: $\gamma = 1$ gives a finite, nonzero boundary slope, $\gamma > 1$ allows inverse demand to be flat, and $0 < \gamma < 1$ allows it to be infinitely steep. The following remark shows that every case delivers the same local size gradients.

Remark 2 (A regular choke price). *Suppose that, for some $c, \gamma > 0$, inverse demand and the derivatives entering the policy formulas satisfy $p(q) = \bar{p} - cq^\gamma + o(q^\gamma)$ as $q \downarrow 0$. With $L(q) := -\mathcal{E}p(q)$, one then has*

$$\begin{aligned} L(q) &\sim \frac{c\gamma}{\bar{p}} q^\gamma, & m(q) &= -(1 + \gamma)L(q) + o(L(q)), \\ \beta_Q(q) &= -\frac{1}{\eta - 1} + \frac{1 + \gamma}{(\eta - 1)^2} L(q) + o(L(q)), & \rho_Q(q) &= -\frac{\gamma(1 + \gamma)}{(\eta - 1)^2} L(q) + o(L(q)), \\ \beta_R(q) &= -\frac{1}{\eta - 1} + \frac{\eta + \gamma}{(\eta - 1)^2} L(q) + o(L(q)), & \rho_R(q) &= -\frac{\gamma(\eta + \gamma)}{(\eta - 1)^2} L(q) + o(L(q)). \end{aligned}$$

Hence the same local slope and curvature gradients obtain. The Klenow–Willis Kimball inverse demand (Klenow and Willis, 2016) satisfies this regular-choke-price condition for every positive Kimball curvature parameter, with regular-choke exponent $\gamma = \nu$.

4.5 Summary and Discussion

We have shown how market power shapes the distribution of firm growth, establishing the size gradient in two ways. First, comparing a price-taking firm with a price-setting firm provides a benchmark: under MSLD*, price setting dampens responses to cost shocks but makes them more asymmetric. Second, we obtain a continuous size gradient among price-setting firms facing a common inverse-demand curve. Near a regular finite choke price, more productive firms operate at larger quantities and have flatter policies with greater curvature magnitude.

Both comparisons apply to output and real revenue. Holding the shock and exposure processes fixed, flatter policies imply lower and less variable growth dispersion, while greater curvature magnitude produces larger skewness fluctuations. The homothetic industry model provides the equilibrium foundation for these results, even though their derivation uses the simpler partial-equilibrium firm problem.

Anti-MSLD* reverses key implications of both comparisons. Relative to a price taker's affine policies, a monopolist has convex policies, reversing the curvature ordering while retaining the dampening effect of price setting on policy slopes. Among monopolists facing the same inverse-demand curve, policies become steeper as operating quantity increases, reversing the slope gradient. Holding shock and exposure processes fixed, these properties imply countercyclical skewness and, within the continuum of monopolists, stronger variance transmission among larger firms. Both implications run counter to the empirical patterns documented above, so we focus on MSLD*.

The model's ingredients, homotheticity, variable-elasticity demand, and its focus on demand structure, place the analysis in the literature on homothetic variable-elasticity demand systems (Feenstra, 2003; Parenti et al., 2017; Matsuyama, 2023). Its central restriction is MSLD*, for which existing evidence on product-level sales reallocation and exporter price–quantity adjustment provides empirical support (Mayer et al., 2021; Berman et al., 2012; Amiti et al., 2019). Like Mrázová and Neary (2017) and Melitz (2018), we express comparative statics through demand elasticities and curvature. By linking MSLD* to cyclical cross-sectional moments of firm growth, the analysis also provides aggregate evidence in support of this demand property.

5 Empirical Evidence

Having shown how heterogeneous exposure and demand curvature can generate the observed size gradient, we now assess this mechanism in three complementary empirical exercises. First, we estimate firm-level responses to identified aggregate shocks and test whether their dispersion and skewness exhibit the opposing size gradients predicted by the model. Second, we use growth moments to estimate relative policy slopes and curvature magnitudes while allowing idiosyncratic risk to differ across size groups. Finally, we examine whether firms with more extreme measured exposures experience larger changes in growth variance and skewness around COVID.

5.1 Data

We use the Compustat sample introduced in Section 2, covering U.S. public firms over 1984Q1–2024Q4. As stated before, real sales are nominal sales deflated by the GDP deflator, and growth is the year-on-year change in log real sales. Firm size is average real sales over the preceding four quarters, and size comparisons are made within two-digit NAICS industries. We require positive reported sales in the quarters needed to construct growth and size, and trim the top and bottom 1 percent of growth observations each quarter. Appendix A details sample construction.

Compustat firms are larger than the representative manufacturing firms in the QFR comparison sample (Appendix Table A.1). Our size comparisons therefore concern heterogeneity within the upper part of the U.S. firm-size distribution: even firms classified as small in our sample need not be small in the broader economy. The following subsections describe additional sample requirements and size classifications used in each exercise.

5.2 Firm-Level Responses to Aggregate Shocks

We use identified aggregate innovations to test the transmission mechanism underlying the size gradient. The model predicts that small firms have steeper real-revenue policies, whereas large firms have more negative policy curvature. Responses to adverse aggregate shocks should therefore be more dispersed among small firms and more negatively skewed among large firms. We examine these predictions using six identified innovations, comparing firm-level response moments across size groups within industries at the horizon of the largest average contraction. This connects the opposing size gradients in realized growth to firms’ transmission of particular aggregate disturbances.

Aggregate innovations and sample. We study innovations to the main business cycle (Angeletos et al., 2020), credit spreads (Gilchrist and Zakrajšek, 2012), oil supply news (Känzig, 2021), monetary policy (Jarociński and Karadi, 2020), investment news (Ben Zeev and Khan, 2015), and financial uncertainty (Ludvigson et al., 2021). Appendix C.1 documents their sources and controls. Each innovation is scaled by its fixed sample standard deviation and oriented so that positive values are growth-contracting. Samples start in 1984Q1, except monetary policy in 1994Q1, and end in 2019Q4, except the main business-cycle series in 2017Q4. Both shock origins and outcomes are restricted to these windows to limit the influence of COVID-19. We retain uninterrupted observation periods containing at least 40 quarters of usable growth for a given firm (a *spell*), allowing several such spells per firm. To classify the spell as a large-firm (small-firm) spell, the firm must remain in the top (bottom) half of its within-industry size distribution throughout the spell. Firm size is measured by average real sales over the preceding four quarters.

Firm-level responses. We allow responses to adverse and favorable innovations to differ, accommodating the asymmetric adjustment predicted by the mechanism. For firm-spell j , shock s , and horizon

$h = 0, \dots, 6$, we estimate the signed local projection (Jordà, 2005)

$$g_{j,t+h} = a_{jh}^s + \tau_{th}^s + b_{jh}^{s,+} x_t^{s,+} + b_{jh}^{s,-} x_t^{s,-} + \sum_{\ell=1}^2 (c_{jh\ell}^s g_{j,t-\ell} + d_{jh\ell}^{s'} \mathbf{C}_{t-\ell}^s) + u_{j,t+h}^s, \quad (28)$$

where $g_{jt} = \log S_{jt} - \log S_{j,t-4}$ is year-on-year real-sales growth, $x_t^{s,+} = \max\{x_t^s, 0\}$, and $x_t^{s,-} = \min\{x_t^s, 0\}$. The vector $\mathbf{C}^s$ contains log real GDP, the log GDP deflator, and the excess bond premium in levels. We include two lags of own growth and each macro variable, allowing all coefficients, including intercepts, to vary by spell. This accommodates heterogeneity in both growth persistence and exposure to aggregate conditions. Time fixed effects τ_{th}^s absorb common movements. The coefficients $b_{jh}^{s,+}$ therefore capture responses to growth-contracting innovations relative to other firms. We study the variance and skewness of these estimated responses.

Distributional comparison. For each shock, we select the horizon h_s^* at which the unweighted mean adverse response is most negative. We compute this mean across all estimated spells using an auxiliary regression with the same regressors and observations but without time fixed effects. This focuses the comparison on the strongest average contraction. As responses dissipate, estimation error can account for an increasing share of their estimated dispersion.

At h_s^* , we retain all adverse coefficients and compute their variance and standardized skewness separately for small and large firms within industry–date cells at adverse-shock dates. We take small-minus-large differences, then average with equal weights across industries within each date and across dates.¹⁸ These differences summarize the size ordering of dispersion and asymmetry: a positive variance difference indicates more dispersed responses among small firms, while a positive skewness difference indicates more negatively skewed responses among large firms. The model implies both differences should be positive.

Coefficient uncertainty and inference. Firm-level shock responses are estimated with error, so dispersion in their estimated coefficients reflects both true response heterogeneity and estimation uncertainty. If small firms' responses are estimated less precisely, this error could contribute to their greater measured dispersion. We therefore adjust coefficient variance by subtracting the estimated contribution of coefficient noise, accounting for covariance between estimation errors across firms.

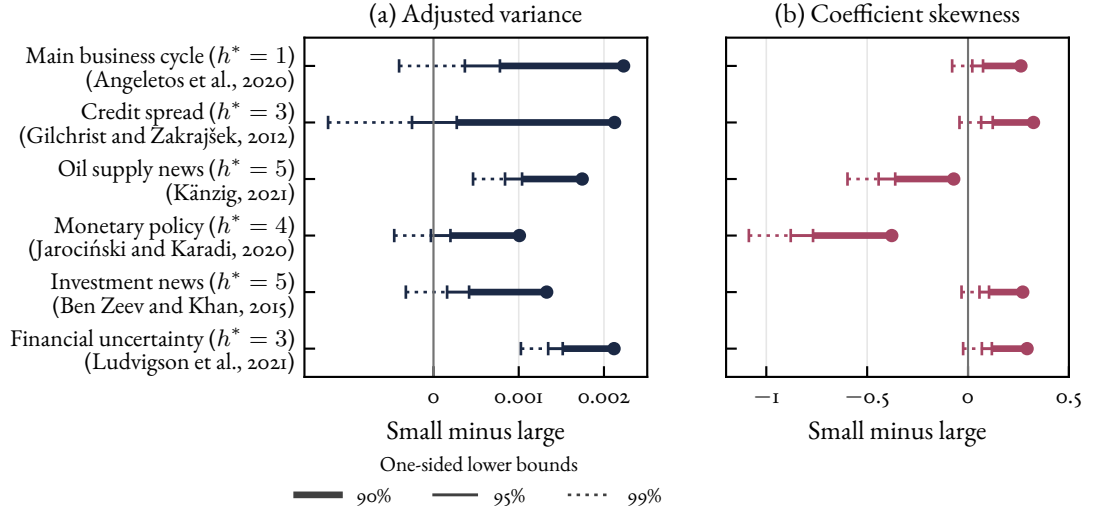
For a cell of n firms, write the estimated coefficient vector as $\hat{\mathbf{b}} = \mathbf{b} + \mathbf{e}$. Let $M_n = I_n - \mathbf{1}_n \mathbf{1}_n' / n$ be the centering matrix and $\hat{\Omega}_e$ the estimated covariance matrix of coefficient errors. The adjusted variance is

$$\hat{V}^{\text{adj}} = \frac{1}{n} \hat{\mathbf{b}}' M_n \hat{\mathbf{b}} - \frac{1}{n} \text{tr}(M_n \hat{\Omega}_e). \quad (29)$$

The first term is observed coefficient variance; the second is its estimated noise component. Appendix

¹⁸Eligible cells contain at least ten stable-small and ten stable-large firms, and eligible dates contain at least five qualifying industries. Cells and aggregation weights remain fixed across bootstrap draws. Appendix C.3 provides the complete construction.

Figure 5: Shock-Mediated Distributional Responses by Firm Size



Note: Small-minus-large coefficient moments at the mean-selected horizon h_s^* . Moments are formed within two-digit NAICS industry–date cells and averaged equally across industries within dates and across dates. Variance is adjusted according to Equation (29); skewness is unadjusted. Bars show one-sided 90, 95 and 99 percent pointwise lower bounds from 1,000 six-quarter block-wild draws; thick, thin and dotted segments indicate the respective confidence levels.

C.3 derives the correction, and Appendix C.5 reports the unadjusted comparison.

Standardized skewness requires a different treatment: coefficient uncertainty matters through both the third moment and the variance used to standardize it. Under centered, symmetric coefficient error independent of the true response, the third moment is unchanged while variance increases. Observed skewness therefore equals $r^{3/2}\Gamma$, where Γ is true response skewness and $r = V(b)/[V(b) + V(e)]$ is coefficient reliability. Comparable reliability across small and large firms thus implies comparable attenuation of their skewness.

We report the skewness of the estimated response coefficients without adjustment. Table C.2 shows nearly identical average attenuation factors across the six shocks: 0.227 for small firms and 0.234 for large firms. The small-firm factor is lower for three shocks and higher for the other three, indicating no systematic size ordering in attenuation. Correcting by a common, fixed attenuation factor of about 0.23 would multiply the point estimates and confidence bounds in Figure 5(b) by approximately 4.3, leaving their signs and statistical conclusions unchanged.

We estimate $\hat{\Omega}_e$ and conduct inference with 1,000 six-quarter calendar-block wild bootstrap draws using OLS residuals. Block multipliers are shared across firms and shocks, with reporting cells fixed across draws. We report one-sided 90, 95 and 99 percent pointwise lower confidence bounds based on the bootstrap standard error (Appendix C.4).

Results. Figure 5 provides the clearest support for the model’s joint ordering under the main business-cycle, investment-news, and uncertainty shocks. For these three shocks, both adjusted-variance and skewness contrasts are positive, with 95 percent lower bounds above zero. The same holds for the adjusted-variance contrast under oil shocks and the skewness contrast under credit shocks. Skewness

contrasts are 0.26, 0.32, 0.27, and 0.29 for the main business-cycle, credit, investment-news, and uncertainty shocks, respectively. The remaining 95 percent lower bounds fall below zero.

The dispersion ordering is also present before the variance correction: all six shocks have positive unadjusted variance contrasts, with 95 percent lower bounds above zero. The correction therefore does not create this ordering. The HAC(8) correction likewise yields positive 95 percent lower bounds for all six shocks (Appendix C.5).

5.3 Estimating Relative Policy Slope, Curvature, and Volatility

Larger firms have lower growth dispersion but greater skewness fluctuations. Lower idiosyncratic risk can amplify standardized skewness fluctuations even when policy curvature is unchanged. We use growth variance and skewness jointly to distinguish differences in firms' policies from differences in their idiosyncratic risk. Throughout, we maintain a common symmetric exposure distribution across size groups and allow idiosyncratic risk to differ across groups but remain constant over time, as discussed in Section 3.3. The curvature index measures the second derivative of the revenue policy relative to its absolute slope. Because our exercise compares the amplitude of adjusted skewness fluctuations, it identifies relative curvature magnitudes, but not their signs.

Identification. We use the baseline within-industry size deciles, pooling firms with the same rank across industries. Let v_{jt} and Γ_{jt} denote growth variance and cumulant skewness in decile j , and let $m_t := \mu_t - \mathbb{E}_{\text{time}}[\mu_t]$ denote mean growth across all firms, demeaned over time. Policy slopes, idiosyncratic volatility, and curvature magnitudes are reported relative to Decile 5.

Policy slopes. We first use fluctuations in growth variance to identify differences in policy responsiveness. At the retained approximation order, time-invariant idiosyncratic risk adds the same component to variance in every period, so it does not affect these fluctuations. With a common exposure distribution and aggregate shock process, their amplitude scales with the squared policy slope. Taking the square root of the ratio of variance fluctuations across groups therefore identifies relative absolute policy slopes, even when idiosyncratic risk differs.

Idiosyncratic risk. We next use variance levels to distinguish greater idiosyncratic risk from steeper policies. Equation (8) gives $\Delta u_t = -m_t/b_m + O(s^2)$, where $b_m > 0$ is the average coefficient on $-\Delta u_t$ in the group mean equations. Substituting this into the variance equation yields $v_{jt} \simeq \alpha_j + \beta_j m_t^2$. The intercept α_j captures the idiosyncratic component of growth variance, while $\beta_j m_t^2$ captures the aggregate component. Both scale with the squared policy slope. Dividing α_j by β_j therefore removes policy responsiveness and gives idiosyncratic variance up to a common scale. Comparing these ratios across groups and taking the square root then identifies relative idiosyncratic volatility. To implement this comparison, we construct α_j and β_j from the time-series moments defined below.

Policy curvature. We then adjust skewness for idiosyncratic risk to identify relative curvature magnitude. To leading order, Equation (10) can be written as

$$\Gamma_{jt} \simeq \rho_{\mathcal{R},j}^* C_\lambda(u_t + u_{t-1}) \text{sgn}(\Delta u_t) A_{jt},$$

where C_λ summarizes how the common exposure distribution amplifies skewness, and A_{jt} is an adjustment factor for idiosyncratic risk, defined in the proposition below. Dividing skewness by A_{jt} leaves a common aggregate factor multiplying each group's curvature. Comparing the root mean squares of these adjusted series on common dates cancels that factor and identifies relative curvature magnitude. Root mean squares measure the size of skewness movements without letting positive and negative values cancel.

Proposition 8 summarizes these relationships. Approximation errors use the local shock scale s .¹⁹

Proposition 8 (Relative identification from observable growth moments). *Suppose the conditions of Proposition 1 and either Proposition 3 or Remark 1 hold groupwise, and assume a common Λ and time-invariant $\sigma_{\kappa,j}^2$. Define*

$$\beta_j := \frac{\text{SD}_{\text{time}}(v_{jt})}{\text{SD}_{\text{time}}(m_t^2)}, \quad \alpha_j := \mathbb{E}_{\text{time}}[v_{jt}] - \beta_j \mathbb{E}_{\text{time}}[m_t^2].$$

Then the following quantities are identified relative to Decile 5:

$$q_j := \frac{|\mathcal{R}'_j(0)|}{|\mathcal{R}'_5(0)|} = \left[\frac{\text{SD}_{\text{time}}(v_{jt})}{\text{SD}_{\text{time}}(v_{5t})} \right]^{1/2} + O(s), \quad (30)$$

$$r_j^\kappa := \frac{\sigma_{\kappa,j}}{\sigma_{\kappa,5}} = \left(\frac{\alpha_j \beta_5}{\beta_j \alpha_5} \right)^{1/2} + O(s), \quad (31)$$

$$r_j^\rho := \frac{|\rho_{\mathcal{R},j}^*|}{|\rho_{\mathcal{R},5}^*|} = \frac{\text{RMS}_{\text{time}}(\Gamma_{jt}/A_{jt})}{\text{RMS}_{\text{time}}(\Gamma_{5t}/A_{5t})} + O(s). \quad (32)$$

Here $\text{RMS}_{\text{time}}(x_t) := \{\mathbb{E}_{\text{time}}[x_t^2]\}^{1/2}$ is the root mean square on the common support, and

$$A_{jt} := \frac{|m_t| (2\alpha_j/\beta_j + \mathcal{K}_\lambda m_t^2)}{\mathcal{K}_\lambda (\alpha_j/\beta_j + m_t^2)^{3/2}}, \quad \mathcal{K}_\lambda := \frac{\text{Var}(\lambda^2)}{\sigma_\lambda^4}.$$

Proof. See Appendix.

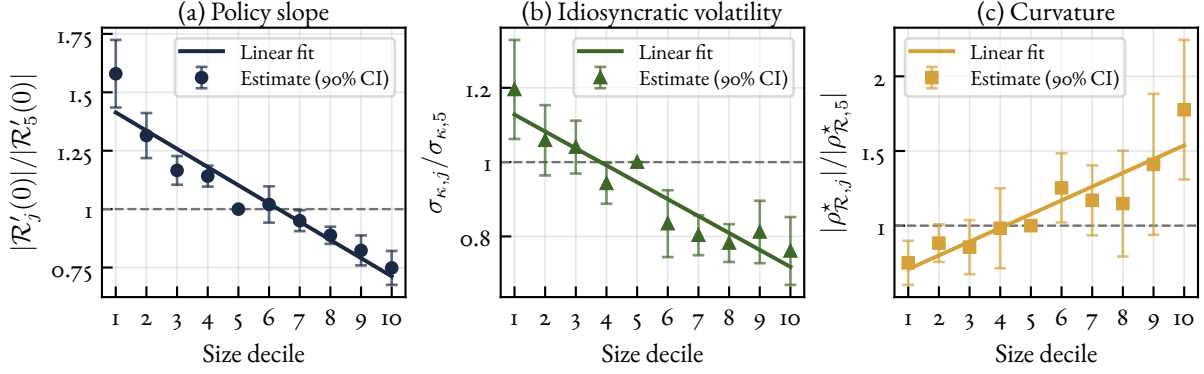
Although the common multiplier C_λ cancels from relative comparisons, identifying relative curvature requires specifying the exposure-shape coefficient $\mathcal{K}_\lambda$, which variance moments do not identify. Our baseline sets $\mathcal{K}_\lambda = 2$, the value implied by $\Lambda = \mathcal{N}(0, \sigma_\lambda^2)$. We assess sensitivity to this choice by varying $\mathcal{K}_\lambda$ over $\{1, 2, 4\}$, which has virtually no effect on our results (see also Appendix D). At $\mathcal{K}_\lambda = 2$, A_{jt}^2 has a transparent interpretation: it is the share of cross-sectional growth variance within size group j attributable to the aggregate shock.²⁰

Estimation. We calculate quarterly moments in each size decile and retain quarters observed for all ten deciles. The baseline uses the 90 percent of these quarters with the smallest m_t^2 , limiting the influence of extreme aggregate-growth observations on the local moment estimates. We estimate the

¹⁹Define $\sigma_{\epsilon,j\tau} := (\sigma_{\kappa,j}^2 + u_\tau^2 \sigma_\lambda^2)^{1/2}$, $s_t := \max_{j,\tau \in \{t-1,t\}} \sigma_{\epsilon,j\tau}$, and $s := \sup_{t \in \mathcal{T}} s_t$ on the common support $\mathcal{T}$.

²⁰Substituting $\mathcal{K}_\lambda = 2$ into the adjustment factor gives $A_{jt}^2 = m_t^2/(\alpha_j/\beta_j + m_t^2) = \beta_j m_t^2/(\alpha_j + \beta_j m_t^2)$. At the retained approximation order, α_j is the idiosyncratic contribution to growth variance and $\beta_j m_t^2$ is the aggregate contribution.

Figure 6: Relative Policy Slopes, Idiosyncratic Volatility, and Curvature Magnitude



Note: Within-industry size deciles; estimates normalized at Decile 5. Bars show 90% delta-method intervals using Newey–West covariance estimates with four calendar-quarter lags; lines are descriptive OLS fits.

quantities in Proposition 8 using their sample counterparts.

For the curvature adjustment only, we smooth the estimated ratios α_j/β_j across deciles using a positive fitted curve; the reported idiosyncratic-volatility estimates use the raw ratios. We also exclude dates on which the adjustment factor is close to zero, since division would magnify small errors in measured skewness.²¹ Confidence intervals use the delta method with Newey–West covariance estimates, conditional on the selected dates and full-sample centering of mean growth. Appendix D details estimation, inference, and sensitivity to the exposure-shape and adjustment-factor choices.

Results. Figure 6 reports estimated policy slopes, idiosyncratic volatility, and curvature magnitudes across size deciles, each relative to Decile 5. Larger firms’ lower growth dispersion reflects both flatter revenue policies and lower idiosyncratic volatility. Panel (a) shows that the relative absolute policy slope falls from 1.58 in Decile 1 to 0.75 in Decile 10, consistent with Proposition 7. Panel (b) shows that relative idiosyncratic volatility declines from 1.20 to 0.76.

The difference in idiosyncratic risk does not account for the full skewness gradient within the maintained model. After adjusting for this difference, relative curvature magnitude rises from 0.75 in the smallest decile to 1.78 in the largest, as shown in panel (c). Larger firms’ greater skewness fluctuations therefore continue to imply greater curvature magnitude after allowing their idiosyncratic risk to be lower. The rank correlations summarize these gradients, although individual decile estimates are not strictly monotone.²²

All three size gradients are also robust across different specifications. They persist when we retain 100, 90, 75, or 50 percent of balanced quarters ranked by m_t^2 , and when we vary the exposure-shape coefficient and adjustment-factor cutoff (Appendix D.1). The gradients also persist with economy-wide

²¹We fit $\hat{\alpha}_j/(2\hat{\beta}_j)$ by nonlinear least squares using $\exp(a + bj + cj^2 + dj^3)$. Comparisons require $A_{jt} \geq 0.05$ in both deciles and at least eight qualifying quarters. Median adjustment factors are well above this cutoff. Profile correlations with the baseline are 0.877, 0.832, and 0.979 at cutoffs of 0.01, 0.025, and 0.075, respectively.

²²Spearman’s correlations and 90% parametric intervals are -0.988 $[-1.000, -0.952]$ for policy slopes, 0.939 $[0.709, 0.976]$ for curvature magnitude, and -0.952 $[-0.976, -0.758]$ for idiosyncratic volatility. Table D.1 reports the full rank-inference results.

size rankings and averages of estimates obtained separately by industry (Appendix D.4).

5.4 Testing the Exposure Mechanism

Proposition 1 assigns heterogeneous exposures a central role in the transmission of aggregate shocks. Holding policy shape and idiosyncratic risk fixed, greater exposure dispersion raises the aggregate contribution to growth variance. Policy curvature also turns this dispersed incidence into asymmetric growth. We now ask whether groups with more dispersed exposures exhibit larger distributional changes during an aggregate contraction, in support of our mechanism.

We study this question using the COVID exposures of Davis et al. (2025). As mentioned above, their measure combines firms' pre-pandemic risk disclosures with equity returns on COVID news days. We compare the outer quintiles (1 and 5) of these signed scores with the middle quintiles (2–4) around the COVID contraction. Pooling both tails targets dispersion in exposure: the outer group contains firms at both extremes of the measured response, whereas the middle group has more similar exposures. To follow the same firms through the COVID contraction and recovery, we require observed annual real-sales growth in every quarter from 2019Q1 to 2021Q1. Exposure groups remain fixed over this window. Within each two-digit NAICS industry, we classify firms as small or large using median pre-event size. The minimum cell-size requirement leaves 906 firms across eight industries.²³

The two exposure groups may nevertheless differ in industry and size composition. Because firm size is linked to policy shape in our framework, these differences can themselves generate different distributional responses. Following Chernozhukov et al. (2013), we therefore give each industry–size cell the same weight in both groups, equal to its share in the combined sample. We then compute mean growth, variance, and skewness from each group's reweighted firm-growth distribution, retaining variation both within and across cells.

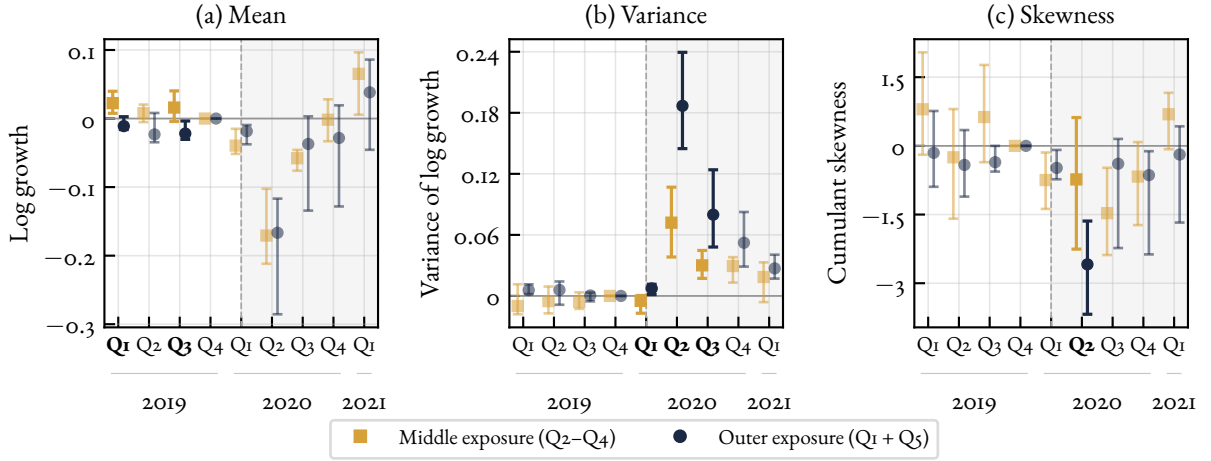
Figure 7 plots each group's moment changes relative to 2019Q4, removing baseline differences in moment levels. Bold quarter labels indicate statistically significant differences between the groups' moment changes. In 2020Q2, mean growth falls by about 0.17 log units in both groups. Their higher moments respond differently: variance rises by 0.187 among outer-exposure firms and 0.072 among middle-exposure firms, while skewness falls by 2.59 and 0.73, respectively. The larger variance increase and larger skewness decline in the outer group are both statistically significant at the 10% level under pointwise inference.²⁴ The larger rise in variance and fall in skewness among firms with more dispersed exposures are consistent with the mechanism's qualitative predictions: heterogeneous exposure spreads the incidence of the contraction, while concave policies make growth more negatively skewed.

We repeat the exercise with inflation exposures in Appendix B.4. In that event setting, the skewness contrast has the opposite sign. However, unlike COVID, the inflation episode lacks a sharply dated onset and overlaps the pandemic recovery and other aggregate news. The exposure groups also display

²³Pre-event size is the prior-four-quarter average of real sales measured at 2019Q4. Each industry–size cell must contain at least eight firms from each exposure group.

²⁴The bootstrap resamples among industries, preserving firm histories and exposure-group masses and recomputing weights and moments.

Figure 7: COVID Exposure and Growth-Distribution Changes



Note: Changes from 2019Q4 at common industry–size shares. Yellow squares: exposure quintiles 2–4; blue circles: 1 and 5. Bars: 90% pointwise percentile intervals from 5,000 industry-pairs bootstrap draws. Bold labels and darker markers and whiskers denote significant outer-minus-middle changes.

substantial pre-event differences in variance and skewness. These features make the inflation episode less informative about the response to a common shock. We therefore place greater weight on the COVID results, which provide clear support for our exposure mechanism.

6 Conclusion

This paper shows how heterogeneous exposure and demand-side nonlinearities allow a common aggregate shock to jointly move mean growth, dispersion, and skewness. The framework also explains why larger firms can have lower and more stable growth dispersion yet larger skewness fluctuations. Evidence from U.S. public firms supports the predicted differences in shock transmission.

These results show how business-cycle moment dynamics can inform the study of demand. Within our framework, the joint behavior of dispersion and skewness provides evidence on the demand properties governing firms' responses to shocks. More generally, higher moments of realized growth distributions depend on both shock incidence and firms' responses. Observed dispersion and skewness therefore do not, by themselves, reveal the shape or evolution of the underlying firm-level shock distribution.

A promising direction for future research is to study the aggregate sources of firm-level shocks and the heterogeneity of firms' exposures to them. Combined with nonlinear responses, simple aggregate disturbances can generate rich distributional dynamics without separately specifying time-varying variances and skewness in primitive shocks. This approach offers a parsimonious foundation for understanding the joint behavior of business-cycle moments.

Appendix

Proof of Proposition 1. We proceed in three steps: expand the fixed policy to second order, calculate the first three cross-sectional moments, and factor the policy coefficients out of the time-series moments. Fix t , condition on $\mathcal{F}_t$, and set

$$X_1 := \epsilon_{it} = \kappa_{it} + u_t \lambda_i, \quad X_0 := \epsilon_{i,t-1} = \kappa_{i,t-1} + u_{t-1} \lambda_i, \quad D := X_1 - X_0, \quad H := X_1^2 - X_0^2.$$

Under the maintained local scaling and moment bounds, $D = O(s_t)$ and $H = O(s_t^2)$ in the norms used below. Consequently, $\mathbb{E}H = O(s_t^2)$, $\text{Var}(D) = \Theta(s_t^2)$ whenever the conditional variance is nondegenerate, and $\text{Cov}(D^2, H) = O(s_t^4)$. Writing $r_1 := \mathcal{R}'(0)$ and $r_2 := \mathcal{R}''(0)$, a second-order expansion gives $\mathcal{R}(X_1) - \mathcal{R}(X_0) = r_1 D + r_2 H/2 + O(s_t^3)$. The common component $a - b_R \Delta u_t + \Delta Z_t$ changes only the mean, while $\Delta \nu_{it} = O_{L^3}(s_t^3)$ contributes beyond the retained orders.

Because $\mathbb{E}\kappa_{it} = \mathbb{E}\lambda_i = 0$, $\mathbb{E}D = 0$, whereas

$$\mathbb{E}H = (\sigma_\kappa^2 + u_t^2 \sigma_\lambda^2) - (\sigma_\kappa^2 + u_{t-1}^2 \sigma_\lambda^2) = \Delta(u_t^2) \sigma_\lambda^2 = O(s_t^2),$$

which proves Equation (8). Moreover,

$$\text{Var}(D) = \text{Var}(\kappa_{it} - \kappa_{i,t-1}) + (\Delta u_t)^2 \text{Var}(\lambda_i) = 2\sigma_\kappa^2 + (\Delta u_t)^2 \sigma_\lambda^2 = \Theta(s_t^2).$$

Only the linear policy term contributes at second order, so $\text{Var}(r_1 D + r_2 H/2) = r_1^2 \text{Var}(D) + O(s_t^3)$, proving Equation (9).

For the leading third central moment, let $H_0 := H - \mathbb{E}H$. Centering and cubing yields

$$\mathbb{E} \left(r_1 D + \frac{r_2}{2} H_0 \right)^3 = r_1^3 \mathbb{E}D^3 + \frac{3}{2} r_1^2 r_2 \mathbb{E}[D^2 H_0] + O(s_t^5).$$

The joint distribution of $(\kappa_{it}, \kappa_{i,t-1}, \lambda_i)$ is invariant under multiplication by -1 , so D is symmetric and $\mathbb{E}D^3 = 0$. Thus

$$\mu_{3,t} = \frac{3}{2} r_1^2 r_2 \text{Cov}(D^2, H) + O(s_t^5). \quad (33)$$

To evaluate the covariance, abbreviate $d := u_t - u_{t-1}$, $s := u_t + u_{t-1}$, $A := \kappa_{it} - \kappa_{i,t-1}$, $K := \kappa_{it}^2 - \kappa_{i,t-1}^2$, and $J := u_t \kappa_{it} - u_{t-1} \kappa_{i,t-1}$. Then $D = A + d\lambda_i$ and $H = K + 2\lambda_i J + sd\lambda_i^2$. Expanding $D^2 = A^2 + 2dA\lambda_i + d^2\lambda_i^2$, independence and symmetry eliminate every pairing whose expectation factors through an odd moment of a centered symmetric variable. For example, $\text{Cov}(A^2, 2\lambda_i J) = 2\mathbb{E}[\lambda_i] \mathbb{E}[A^2 J] = 0$, while $\text{Cov}(A^2, sd\lambda_i^2) = 0$ by independence. The surviving terms are

$$\text{Cov}(D^2, H) = \text{Cov}(A^2, K) + \text{Cov}(2dA\lambda_i, 2\lambda_i J) + \text{Cov}(d^2\lambda_i^2, sd\lambda_i^2).$$

The first is zero because the two iid idiosyncratic shocks enter symmetrically. Since $\mathbb{E}[AJ] = (u_t + u_{t-1})\sigma_\kappa^2 = s\sigma_\kappa^2$, the second is $4ds\sigma_\kappa^2\sigma_\lambda^2$; the third is $sd^3\text{Var}(\lambda^2)$. Hence

$$\text{Cov}(D^2, H) = 4(\Delta u_t)(u_t + u_{t-1})\sigma_\kappa^2\sigma_\lambda^2 + \Delta(u_t^2)(\Delta u_t)^2 \text{Var}(\lambda^2) = O(s_t^4).$$

Let $V_{D,t} := 2\sigma_\kappa^2 + d^2\sigma_\lambda^2 = \Theta(s_t^2)$. The variance expansion implies

$$v_t^{3/2} = |r_1|^3 V_{D,t}^{3/2} + O(s_t^4).$$

Since $\text{Cov}(D^2, H) = O(s_t^4)$, dividing Equation (33) by $v_t^{3/2}$ and using $r_2/|r_1| = \rho_{\mathcal{R}}^*$ yields Equation (10), including its $O(s_t^2)$ remainder.

Finally write $v_t = r_1^2 V_t$ and $\Gamma_t = \rho_{\mathcal{R}}^* G_t$, where V_t and G_t depend only on the fixed shock and exposure processes. Stationarity and ergodicity give $\mathbb{E}_{\text{time}}[v_t] = r_1^2 \mathbb{E}_{\text{time}}[V_t]$ and $\text{SD}_{\text{time}}(v_t) = r_1^2 \text{SD}_{\text{time}}(V_t)$, with analogous relations involving $\rho_{\mathcal{R}}^*$ and $|\rho_{\mathcal{R}}^*|$ for the mean and SD of Γ_t . This proves the proportionalities whenever the corresponding common time moment is nonzero. Replacing $\mathcal{R}$ by $\mathcal{Q}$ and setting $\zeta \equiv 0$ proves the output version. $\blacksquare$

Proof of Proposition 3. We express the price-forecast error as a smooth function of own and rival log-output deviations. A second-order expansion in rivals' deviations depends only on their cross-sectional mean and variance; symmetry makes both common across firms at second order, while interactions with the firm's own first-order deviation are third order.

Fix t and define the shock scale $s := \sigma_{\epsilon,t} = (\sigma_{\kappa}^2 + u_t^2 \sigma_{\lambda}^2)^{1/2}$. All order statements below refer to $s \downarrow 0$. Symmetry gives a common benchmark $\bar{q}$, policy $x(\epsilon)$, and log policy $\mathcal{Q}(\epsilon) = \log x(\epsilon)$. Define $\delta_{it} := \log x(\epsilon_{it}) - \log \bar{q}$. Bounded standardized shocks and smoothness imply $\epsilon_{it} = O(s)$ and $\delta_{it} = O(s)$ uniformly. Taylor expansion gives

$$\delta(e) = \mathcal{Q}'(0)e + \frac{1}{2}\mathcal{Q}''(0)e^2 + O(e^3).$$

Since $\mathbb{E}e = 0$, $\mathbb{E}e^2 = \sigma_{\epsilon,t}^2$, and $\mathbb{E}|e|^3 = O(s^3)$, the cross-sectional mean and variance of rival deviations satisfy

$$\begin{aligned} M_t &:= \mathbb{E}_i \delta_{it} = \frac{1}{2}\mathcal{Q}''(0)\sigma_{\epsilon,t}^2 + O(s^3) = O(s^2), \\ V_t &:= \text{Var}_i(\delta_{it}) = \mathcal{Q}'(0)^2 \sigma_{\epsilon,t}^2 + O(s^3) = O(s^2). \end{aligned}$$

To make the price-forecast error explicit, approximate the continuum by N symmetric rivals. For rival deviations z_j , define

$$G_N(\delta, z_1, \dots, z_N) := \log h_i(\bar{q}e^{\delta}, \{\bar{q}e^{z_j}\}_{j=1}^N).$$

The first argument is $\bar{q}e^{\delta}$ because $\delta = \log q_i - \log \bar{q}$. The exact finite- N forecast error is

$$\zeta_{i,N} = G_N(\delta_{it}, \mathbf{z}_t) - G_N(\delta_{it}, \mathbf{0}) :$$

the first term evaluates the realized relative-price derivative at actual rival quantities, while the second holds every rival at $\bar{q}$. At $\mathbf{z} = \mathbf{0}$, a second-order Taylor expansion gives

$$G_N(\delta, \mathbf{z}) - G_N(\delta, \mathbf{0}) = \sum_j G_j(\delta) z_j + \frac{1}{2} \sum_j \sum_k G_{jk}(\delta) z_j z_k + O(\|\mathbf{z}\|^3).$$

Symmetry across rivals implies $G_j = a_N$, $G_{jj} = b_N$, and $G_{jk} = c_N$ for $j \neq k$, with coefficients depending on δ . Let $\bar{z}_N = N^{-1} \sum_j z_j$ and $\bar{z}_N^2 = N^{-1} \sum_j z_j^2$. Because

$$\sum_{j \neq k} z_j z_k = N^2 \bar{z}_N^2 - N \bar{z}_N^2,$$

the expansion becomes

$$G_N(\delta, \mathbf{z}) - G_N(\delta, \mathbf{0}) = A_N(\delta) \bar{z}_N + B_N(\delta) \bar{z}_N^2 + C_N(\delta) \bar{z}_N^2 + O(\|\mathbf{z}\|^3),$$

where $A_N = N a_N$, $B_N = N(b_N - c_N)/2$, and $C_N = N^2 c_N/2$.

Assuming these coefficient combinations and the third-order remainder have finite uniform local

continuum limits, the law of large numbers gives $\bar{z}_N \rightarrow M_t$ and $\bar{z}_N^2 \rightarrow V_t + M_t^2$. Since $M_t^2 = O(s^4)$,

$$\zeta_{it} = A(\delta_{it})M_t + B(\delta_{it})V_t + O(s^3).$$

Smoothness in own deviation implies $A(\delta_{it}) = A_0 + A_1\delta_{it} + O(\delta_{it}^2)$ and analogously for B . Because $\delta_{it} = O(s)$ while $M_t, V_t = O(s^2)$, every term involving $\delta_{it}M_t$ or $\delta_{it}V_t$ is third order. Therefore

$$\zeta_{it} = \left[\frac{1}{2}A_0\mathcal{Q}''(0) + B_0\mathcal{Q}'(0)^2 \right] \sigma_{\epsilon,t}^2 + O(s^3).$$

Define $\chi_h := A_0\mathcal{Q}''(0)/2 + B_0\mathcal{Q}'(0)^2$ and $Z_t := \chi_h(\sigma_\kappa^2 + u_t^2\sigma_\lambda^2)$. Then $\zeta_{it} = Z_t + \nu_{it}$ with $\sup_i |\nu_{it}| = O(s^3)$, so $\|\nu_t\|_{L^3(i)} \leq \sup_i |\nu_{it}| = O(s^3)$. Also $\Delta Z_t = \chi_h\sigma_\lambda^2\Delta(u_t^2)$. ■

Proof of Lemma 1. We first compare marginal revenue with marginal cost and then invert the log first-order condition. Differentiating profits gives $\pi'(q) = R'(q) - \eta\phi e^\epsilon q^{\eta-1}$. Marginal revenue is strictly decreasing and marginal cost strictly increasing, so the maintained interior crossing is unique and satisfies Equation (18). Taking logs gives $\epsilon = H(\hat{q})$, with

$$H'(\hat{q}) = \frac{e^{\hat{q}}R''(e^{\hat{q}})}{R'(e^{\hat{q}})} - (\eta - 1) = m(e^{\hat{q}}) - (\eta - 1) < 0.$$

Hence the inverse policy $\mathcal{Q} = H^{-1}$ is decreasing. Finally, $R'(q) = p(q)[1 + \mathcal{E}p(q)] > 0$ and $p'(q) < 0$ imply $\mathcal{E}p(q) \in (-1, 0)$, while dividing price by marginal cost gives $\mu(q) = [1 + \mathcal{E}p(q)]^{-1}$. ■

Lemma 2 (Local Curvature of Output). *Let $\hat{q}^* := \mathcal{Q}(\epsilon^*)$ and $q^* := e^{\hat{q}^*}$. If $H'(\hat{q}^*) \neq 0$, then*

$$\mathcal{Q}'(\epsilon^*) = \frac{1}{H'(\hat{q}^*)}, \quad \mathcal{Q}''(\epsilon^*) = -\frac{H''(\hat{q}^*)}{H'(\hat{q}^*)^3}, \quad \rho_{\mathcal{Q}}^* = \frac{H''(\hat{q}^*)}{H'(\hat{q}^*)^2}, \quad (34)$$

$$H'(\hat{q}^*) = m(q^*) - (\eta - 1), \quad H''(\hat{q}^*) = \left. \frac{dm(q)}{d \log q} \right|_{q=q^*}. \quad (35)$$

Thus $m'(q^*) < 0$ is equivalent to $\mathcal{Q}''(\epsilon^*) < 0$ and $\rho_{\mathcal{Q}}^* < 0$. In particular, $MSLD^*$ implies local concavity, whereas *anti-MSLD** implies local convexity.

Proof of Lemma 2. Our argument differentiates the inverse identity $H(\mathcal{Q}(\epsilon)) = \epsilon$. Once and twice differentiation give

$$H'(\mathcal{Q})\mathcal{Q}' = 1, \quad H''(\mathcal{Q})(\mathcal{Q}')^2 + H'(\mathcal{Q})\mathcal{Q}'' = 0,$$

and hence $\mathcal{Q}' = 1/H'$ and $\mathcal{Q}'' = -H''/H'^3$. At the optimum, $m(q^*) < 0$ and $\eta > 1$, so $H'(\hat{q}^*) < 0$ and $|\mathcal{Q}'| = 1/H'$. Therefore $\rho_{\mathcal{Q}}^* = H''/H'^2$.

Differentiating the definition of H yields

$$H'(\hat{q}) = \frac{e^{\hat{q}}R''(e^{\hat{q}})}{R'(e^{\hat{q}})} - (\eta - 1) = m(e^{\hat{q}}) - (\eta - 1), \quad H''(\hat{q}) = e^{\hat{q}}m'(e^{\hat{q}}).$$

Thus $H''(\hat{q}^*) = q^*m'(q^*)$ has the sign of $m'(q^*)$. The formulas for $\mathcal{Q}''$ and $\rho_{\mathcal{Q}}^*$ give the stated equivalence and the curvature implications of $MSLD^*$ and *anti-MSLD**. ■

Proof of Corollary 2. Specializing Proposition 1 to $\sigma_\kappa^2 = 0$ and evaluating the prescribed shock path gives the result. With $C_\lambda := (3/2)\text{Var}(\lambda^2)/\sigma_\lambda^3 > 0$, the skewness formula becomes

$$\gamma_1(g_t^Y) \simeq \rho_Y^* C_\lambda (u_t + u_{t-1}) \text{sgn}(\Delta u_t).$$

On impact, $u_0 = 0$ and $u_1 = x$, so $\gamma_1(g_1^Y) = \rho_Y^* C_\lambda x < 0$. For $t \geq 2$, $u_t = \beta^{t-1}x$, $u_{t-1} = \beta^{t-2}x$, and $\Delta u_t = -(1 - \beta)\beta^{t-2}x < 0$. Hence

$$\gamma_1(g_t^Y) = -\rho_Y^* C_\lambda (1 + \beta)\beta^{t-2}x > 0, \quad \frac{|\gamma_1(g_{t+1}^Y)|}{|\gamma_1(g_t^Y)|} = \beta.$$

The simplified expression is linear in both ρ_Y^* and x , proving the amplitude statements. $\blacksquare$

Proof of Proposition 7. We combine the output-policy expansion in Proposition 6 with the local inverse-demand elasticity. Its boundary calculation gives $p'(0) = -\iota_{R'}\bar{p}/2$, and therefore

$$p(q) = \bar{p} - \frac{\iota_{R'}\bar{p}}{2}q + O(q^2), \quad \varepsilon(q) = \frac{qp'(q)}{p(q)} = -\frac{\iota_{R'}}{2}q + O(q^2).$$

Because $\mathcal{R} = \log[p(q)q]$ composed with the output policy, $\beta_R(q) = [1 + \varepsilon(q)]\beta_Q(q)$. Set $A := \eta - 1$. Proposition 6 then gives

$$\beta_R(q) = \left[1 - \frac{\iota_{R'}}{2}q + O(q^2)\right] \left[-\frac{1}{A} + \frac{\iota_{R'}}{A^2}q + O(q^2)\right] = -\frac{1}{\eta - 1} + \frac{\iota_{R'}}{2} \frac{\eta + 1}{(\eta - 1)^2}q + O(q^2).$$

For curvature, Proposition 5 gives $\rho_R = \rho_Q + \varepsilon\psi|\beta_Q|/(1 + \varepsilon)$. Writing $\ell := \iota_{R'}/2$, the expansion $\varepsilon(q) = -\ell q + O(q^2)$ implies $\psi(q) = q\varepsilon'(q)/\varepsilon(q) = 1 + O(q)$. Also $|\beta_Q| = A^{-1} + O(q)$ and $1 + \varepsilon = 1 + O(q)$. Hence

$$\rho_R(q) = -\frac{\iota_{R'}}{A^2}q - \frac{\iota_{R'}}{2A}q + O(q^2) = -\frac{\iota_{R'}}{2} \frac{\eta + 1}{(\eta - 1)^2}q + O(q^2).$$

Differentiating gives the stated local signs, exactly as in Proposition 6. Finally, $d[p(q)q]/dq = R'(q) > 0$ locally, so revenue preserves the quantity ranking. $\blacksquare$

Proof of Proposition 8. The proof proceeds in three stages: use mean growth as a local proxy for the aggregate shock, identify relative slopes and idiosyncratic volatility from the affine variance equation, and remove the resulting group-specific attenuation from skewness before taking RMS ratios.

Average Proposition 1's group mean equation across the ten equal-mass deciles. Demeaning over time removes constants and, under stationarity, the time means of Δu_t , $\Delta(u_t^2)$, and the common residual difference. Since $\Delta(u_t^2) = (u_t + u_{t-1})\Delta u_t$ and both factors are first order in s_t ,

$$m_t = -b_m \Delta u_t + \frac{\overline{\mathcal{R}''(0)}}{2} \sigma_\lambda^2 (u_t + u_{t-1}) \Delta u_t + \Delta \bar{Z}_t + O(s_t^3) = -b_m \Delta u_t + O(s_t^2).$$

Local inversion gives $\Delta u_t = -m_t/b_m + O(s_t^2)$. Squaring yields $(\Delta u_t)^2 = m_t^2/b_m^2 + O(s_t^3)$, while multiplication by $u_t + u_{t-1}$ gives $(u_t + u_{t-1})\Delta u_t = (u_t + u_{t-1}) \text{sgn}(\Delta u_t)|m_t|/b_m + O(s_t^3)$. The square has a third-order remainder because it multiplies a first-order leading term by a second-order inversion error.

Within group j , set $r_j := \mathcal{R}'_j(0)$. Proposition 1 gives $v_{jt} = r_j^2[2\sigma_{\kappa,j}^2 + \sigma_\lambda^2(\Delta u_t)^2] + O(s_t^3)$. Define $B_j := r_j^2\sigma_\lambda^2/b_m^2$ and $I_j := 2r_j^2\sigma_{\kappa,j}^2$. Substitution yields $v_{jt} = I_j + B_j m_t^2 + O(s_t^3)$. Taking time-series standard deviations removes I_j , so $\text{SD}_{\text{time}}(v_{jt}) = B_j \text{SD}_{\text{time}}(m_t^2) + O(s^3)$ and $\beta_j = B_j + O(s)$. Consequently, $[\text{SD}_{\text{time}}(v_{jt})/\text{SD}_{\text{time}}(v_{5t})]^{1/2} = (B_j/B_5)^{1/2} + O(s) = |r_j|/|r_5| + O(s)$. Taking time means of the affine representation and subtracting $\beta_j \mathbb{E}_{\text{time}}(m_t^2)$ gives $\alpha_j = I_j + O(s^3)$ and hence $\alpha_j/(2\beta_j) = b_m^2 \sigma_{\kappa,j}^2 / \sigma_\lambda^2 + O(s^3)$. The ratio with Decile 5 followed by the positive square root proves the relative-volatility formula.

It remains to verify the nonlinear attenuation correction. Proposition 1 gives

$$\Gamma_{jt} = \rho_j \frac{3}{2} \frac{(u_t + u_{t-1}) \Delta u_t [4\sigma_{\kappa,j}^2 \sigma_\lambda^2 + (\Delta u_t)^2 \text{Var}(\lambda^2)]}{[2\sigma_{\kappa,j}^2 + (\Delta u_t)^2 \sigma_\lambda^2]^{3/2}} + O(s_t^2).$$

Let $K := \text{Var}(\lambda^2)$ and $\ell := \sigma_\lambda^2$, treating b_m, K, ℓ as positive order-one constants. Then $\Delta u_t = O(s)$, $\sigma_{\kappa,j}^2 = O(s^2)$, $|m_t| = b_m |\Delta u_t| + O(s^2)$, $m_t^2 = b_m^2 (\Delta u_t)^2 + O(s^3)$, and $\alpha_j/\beta_j = 2b_m^2 \sigma_{\kappa,j}^2/\ell + O(s^3)$.

Write the numerator and denominator of A_{jt} as $N_{jt} := |m_t| [2\alpha_j/\beta_j + (K/\ell^2)m_t^2]$ and $D_{jt} := (K/\ell^2)(\alpha_j/\beta_j + m_t^2)^{3/2}$. The bracket in N_{jt} equals $(b_m^2/\ell^2)[4\sigma_{\kappa,j}^2 \ell + K(\Delta u_t)^2] + O(s^3) = O(s^2)$. Multiplication by $|m_t|$ therefore gives

$$N_{jt} = \frac{b_m^3 |\Delta u_t|}{\ell^2} [4\sigma_{\kappa,j}^2 \ell + K(\Delta u_t)^2] + O(s^4) =: N_{jt}^0 + O(s^4), \quad N_{jt}^0 = O(s^3).$$

For the denominator, $X_{jt} := \alpha_j/\beta_j + m_t^2 = (b_m^2/\ell)[2\sigma_{\kappa,j}^2 + \ell(\Delta u_t)^2] + O(s^3) =: X_{jt}^0 + O(s^3)$. Assume the leading variance mixture is nondegenerate, so $X_{jt}^0 = \Theta(s^2)$. Taylor expansion of $x^{3/2}$ around X_{jt}^0 gives $(X_{jt})^{3/2} = (X_{jt}^0)^{3/2} + O(s^4)$, and therefore

$$D_{jt} = \frac{K b_m^3}{\ell^{7/2}} [2\sigma_{\kappa,j}^2 + \ell(\Delta u_t)^2]^{3/2} + O(s^4) =: D_{jt}^0 + O(s^4), \quad D_{jt}^0 = \Theta(s^3).$$

Dividing the two expansions yields

$$A_{jt} = \frac{|\Delta u_t| \ell^{3/2}}{K} \frac{4\sigma_{\kappa,j}^2 \ell + K(\Delta u_t)^2}{[2\sigma_{\kappa,j}^2 + \ell(\Delta u_t)^2]^{3/2}} + O(s).$$

The fourth-order input errors become first order because the denominator is order s^3 . Set $C_\lambda := (3/2)K/\ell^{3/2}$. Since $u_t + u_{t-1} = O(s)$, the first-order error in A_{jt} becomes second order after multiplication by $u_t + u_{t-1}$. Comparing the preceding expression with the skewness equation gives $\Gamma_{jt} = \rho_j C_\lambda (u_t + u_{t-1}) \text{sgn}(\Delta u_t) A_{jt} + O(s_t^2)$. If A_{jt} is bounded away from zero, division is regular and $\Gamma_{jt}/A_{jt} = \rho_j C_\lambda (u_t + u_{t-1}) \text{sgn}(\Delta u_t) + O(s_t^2)$. On common dates, the time series multiplying ρ_j is identical across groups. The reverse triangle inequality for the RMS norm, $|\text{RMS}(X + R) - \text{RMS}(X)| \leq \text{RMS}(R)$, preserves the uniform second-order remainder. Therefore

$$\text{RMS}_{\text{time}}(\Gamma_{jt}/A_{jt}) = |\rho_j| C_\lambda \text{RMS}_{\text{time}}[(u_t + u_{t-1}) \text{sgn}(\Delta u_t)] + O(s^2).$$

If the common RMS factor is $\Theta(s)$ and $\rho_5 \neq 0$, taking the ratio with Decile 5 proves the relative-curvature formula with remainder $O(s)$. ■

Proposition 9 (CES as a knife-edge benchmark). *If $p(q) = Aq^{-\chi}$ with $A > 0$ and $\chi \in (0, 1)$, then $\varepsilon(q) \equiv m(q) \equiv -\chi$ and $\psi(q) \equiv 0$. Consequently, $\rho_Q^* = \rho_R^* = 0$ at every operating quantity. CES therefore remains the knife-edge case in which the output policy is locally linear, the direct real-revenue-curvature term vanishes, and skewness is absent to second order.*

Proof of Proposition 9. Under CES, both the marginal-revenue elasticity and the log first-order condition are affine. Differentiating $p(q) = Aq^{-\chi}$ gives $\varepsilon(q) = qp'(q)/p(q) = -\chi$, hence $\psi(q) = 0$. Moreover,

$$R'(q) = p(q) + qp'(q) = A(1 - \chi)q^{-\chi}, \quad m(q) = \frac{qR''(q)}{R'(q)} = -\chi.$$

The log first-order condition is therefore

$$\epsilon = \log A + \log(1 - \chi) - \log \eta - \log \phi - (\eta - 1 + \chi)\widehat{q},$$

so its inverse $\mathcal{Q}(\epsilon)$ is affine and $\rho_Q^* = 0$. Equation (23) then gives $\rho_R^* = 0$ because both ρ_Q^* and ψ vanish. Proposition 1's leading skewness term is proportional to the relevant curvature index and therefore vanishes for output and real sales. ■

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